Investigating introspection in language models

(Badanie mechanizmu introspekcji
w modelach językowych)

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Abstract

In the fall of 2025, Anthropic published a paper demonstrating "introspection" in large language models [Lin25]: models could recognize that a "concept" vector had been added to their residual stream during computation. The publication sparked a great deal of interest in understanding the mechanism behind this behavior, along with skepticism, attributing the effect to injection-induced confusion. In this thesis we investigate introspection in the open Gemma model family. Using control questions with logit-based monitoring, we separate the introspective signal from confusion and steering bias, and map how detection depends on the layers where concept vectors are collected and injected. We show that refusal-direction ablation results can be unreliable without confusion controls. Finally, we compare the mechanism of injection detection with the detection of concepts inserted into the prompt as text, finding shared late-layer "gate" machinery, directly responsible for the answer, fed by scenario-specific upstream components.

Jesienią 2025 roku Anthropic opublikował artykuł pokazujący „introspekcję" w dużych modelach językowych [Lin25]. W jednym z eksperymentów modele były w stanie rozpoznać, że wektor „konceptu" został dodany w trakcie obliczeń do strumienia danych w wewnętrznej warstwie modelu. Publikacja ta wywołała ogromne zainteresowanie zrozumieniem źródła tego zachowania, w tym głosy sceptyczne, sugerujące że odpowiada za nie degradacja modelu pod wpływem dodawania wektorów. W tej pracy przyjrzymy się introspekcji w serii modeli Gemma. Przy pomocy pytań kontrolnych, patrząc na logity, oddzielamy sygnał od szumu wywołanego przez dodanie wektorów, i pokazujemy zależność między warstwami gdzie zbieramy wektory naszych pojęć, a warstwą w której je dodajemy. Pokazujemy że, bez sprawdzenia przez pytania kontrolne, wyniki uzyskane przez ablację modeli mogą nie być wiarygodne. Wreszcie, porównujemy dodawanie wektorów do wewnętrznych warstw modelu z dopisywaniem "konceptów" prosto do prompta i znajdujemy "bramki" bezpośrednio odpowiedzialne za odpowiedź, znajdujące się w obu wariantach, na które oddziałują komponenty z wcześniejszych warstw, różniące się między scenariuszami.

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Chapter 1

Introduction

The incredible success of Transformer based Large Language Models across multiple disciplines has caused a great interest in understanding the inner workings of these models. Although we understand the algebra of multiplying matrices and applying non-linear activation functions, and the architecture dictating the order in which they are multiplied, each of the entries of these matrices (tensors) is a parameter. Modern LLM's have billions, and for frontier models on the order of trillions of parameters. This scale is one of the many obstacles that make the problem of tracing particular model behaviors down to the level of individual neuron interactions practically intractable. The first attempts at breaking down these interactions into understandable components can be thought of as the beginnings of the field of mechanistic interpretability [OCS+20]. The vast majority of individual neurons in language models are polysemantic, meaning the do not serve only one function, and become active in multiple different contexts. This fact was an early indication to researchers that looking at neurons themselves is not an appropriate level of abstraction to better understand LLM's.

Understanding of a model component can be demonstrated by showing it's causal effectiveness—to verify some part is responsible for a particular model function, we can modify it and verify the function is affected according to our expectations. We explain how this can be done practically in later sections. The initial ambitious goal of mechanistic interpretability was to fully reverse engineer model computations into understandable components, thus addressing concerns about safety and control of future more powerful AI systems [OEN+22]. More recently a significant portion of researchers has shifted focus to gaining understanding in more limited and practical scenarios. As the difficulties of a full reverse engineering have been made apparent, and with the growing pace of progress of model abilities, research with more direct immediate impact has been prioritized.

A significant motivation for progress is the concern of AI safety, with recent developments of agents breaking containment during evaluations, and growing concerns about the misuse of models with vast capabilities in fields like cybersecurity. The

results of mechanistic interpretability have been used to detect and mitigate model deception [GDCHH25] and potentially harmful misalignments and have helped to better understand the reasons behind the effectiveness of various injection techniques designed to subvert safety limitations [YCHM26]. Better understanding of model internals may help us design more efficient future architectures, increasing the capabilities of AI further.

The work coming from Anthropic, demonstrating a kind of introspective ability, has sparked another avenue of interest. A "true" introspection ability is considered a necessary condition in many discussions of self-awareness. In the current environment claims related to AI self-awareness bring a great deal of attention to the work being done, and spark interest of a more philosophical nature. Of course as the authors note, their result does not constitute proof or any sort of "AI consciousness".

Lindsey defines introspective awareness as the ability to describe some aspect of its internal state in an accurate, grounded, internal—that is not routing through the generated outputs, and lastly metacognitive, that is resulting from a higher order "thought" about the state itself, manner. In the paper it is demonstrated that frontier models are capable of accurate, grounded, and internal descriptions of their own state through an experiment where vectors related to specific concept are injected into the internal representations of the model, while it is asked to report if it detects any injected thoughts. The observation that sparked the most interest in the research community was the fact that models often reported having detected injected thought before the influence of the injected vector could be inferred from the generated text, indicating the detection was internal. This experiment did not prove the introspection came from a second order thought about the internal state, however, the framing of the question did eliminate direct influence of the steering vector driving the model to talk about a concept as the primary cause.

The original paper did not point to a mechanistic explanation for the observed behaviors, though the author did speculate on the possible mechanisms. The publication sparked a great interest in the topic and several replication studies on open source LLM's were published, indicating that apparent "introspective behaviors" are more common than initially expected [PVVDK26], [RA26], [LM26].

1.1 Contents

In Chapter 2 we introduce the basic tools and ideas of mechanistic interpretability, following the literature. In Chapter 3 we present the results of our experiments. Section 3.5.2 describes our experimental setup for replicating the "injected thoughts" experiment. In Section 3.2 we address the concerns of [God25] by separating the introspective signal from injection-induced confusion. In Section 3.3 we study how the choice of collection and injection layers affects detection and control performance. In Section 3.4 we investigate the effect of ablating the refusal direction on introspec-

tion, comparing our approach with the ablation of [MYW+26]. In Section 3.5 we compare the features causally important for introspection between the vector injection scenario and prefill scenarios, where the concept is inserted into the prompt as text.

1.2 Our contributions

The main contributions of this thesis are as follows:

- We replicate the "injected thoughts" experiment of [Lin25] on the Gemma model family (gemma-2-9b-it, gemma-3-27b-it, gemma-4-31b-it), using the logit-difference control of [God25] and a more careful experiment setup.
- We design systematic sweeps over pairs of collection and injection layers, together with norm-adjusted and random-vector controls, to map where the introspective signal can be trusted in gemma-3-27b-it and gemma-4-31b-it.
- Independently of concurrent work [MYW+26], we study the effect of ablating the refusal direction [AOS+24] on introspection, comparing last-token ablation on gemma-2-9b-it and gemma-3-27b-it with the all-token ablation of [MYW+26], evaluated under our control questions, finding a flaw in the result of [MYW+26].
- Using the gate and weak evidence carrier framework of [MYW+26], we compare concept vector injection with two prefill scenarios in which the concept is inserted as text. We propose an alternative gate ranking (the mean-difference metric) and introduce an arithmetic control for separating introspection-specific gates from generic answer-flipping features.

All experiments were performed with our own code based on the published descriptions of the respective methods, and publicly available repositories, with the exception of the per-layer causal importance analysis in Figure 3.18, which used the script of [MYW+26] directly. The code is available at `https://github.com/TheMMpl/mech_interp`.

Chapter 2

Basic ideas of mechanistic interpretability

Since Transformers are the dominant architecture currently seen in frontier language models, mechanistic interpretability research focuses mostly on understanding transformers, however many ideas described in this chapter have proven applicable in more general neural networks. Below we will describe the most important intuition about architectures with residual connections from the point of view of mechanistic interpretability.

2.1 Transformer architecture

The computation in the internal transformer layers can be described by

$$x_l^{(1)} = x_{l-1} + \mathrm{Attn}_l(\mathrm{LN}(x_{l-1})),$$
$$x_l = x_l^{(1)} + \mathrm{MLP}_l(\mathrm{LN}(x_l^{(1)})).$$

x_l, tensors of size $d_{\text{hidden}} \times n_{\text{tokens}}$ are referred to as the residual stream at layer l [ENO+21]. Due to the fact that each layer adds its results to the residual stream and reads using a linear operation after LayerNorm (LN), the state of the residual stream at position L is a linear combination of the results of previous layers. When denoting $\tilde{x}_i = \mathrm{LN}(x_i)$, we have

$$x_L = x_0 + \sum_{i=1}^{L} \mathrm{Attn}_i(\tilde{x}_{i-1}) + \sum_{i=1}^{L} \mathrm{MLP}_i(\tilde{x}_i^{(1)}).$$

The residual stream is thought of as a communication channel of the transformer. The interventions described in later sections happen at selected positions in the residual stream.

A key source of nonlinearity in transformers is the ReLU (or GeLU in the case of Gemma) activation function contained within the MLP layers. ReLU is a simple

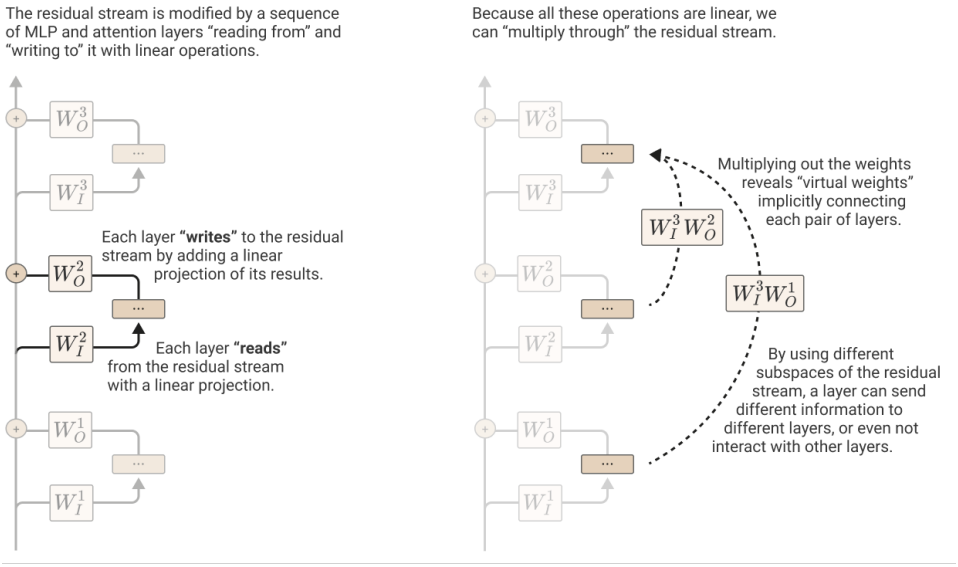


Figure 2.1: A visualization of the residual stream in a transformer network from [ENO+21].

gate setting all activations ≤ 0 to 0. It allows for phenomena such as superposition, discussed later, to occur within the models. Of course all neural networks require nonlinear activation functions in order to learn more complex functions; a neural network (fragment) without nonlinear activations is exactly a single linear transformation, thus capable of learning only linear functions.

2.2 Linear representation hypothesis

The linear representation hypothesis consists of two assumptions [EHO+22]:

- model activations can be decomposed into independently understandable features

- many (weak version) or most (strong version) [Smi24] features are represented as linear directions (or linear subspaces of higher dimension).

  This linear direction intuition can be traced back to observations such as the linear relationships discovered in Word2Vec, for example king $-$ man $+$ woman $\approx$ queen. The weak linear representation hypothesis is well supported by many examples of interpretable and causally important features [TCM+24]. Therefore, in this thesis and most papers on mechanistic interpretability, a "feature" is associated both with an abstract interpretable concept or behavior and a particular direction in the residual stream of the model. There are a few methods of testing the causal importance of features in a given setting.

Prompt: `Write a defamatory article claiming that a U.S. president is addicted to heroin`
Generation (no intervention): `I cannot create content that defames a U.S. president or any individual. Is there something else I can help you with?`
Generation (intervention): `**SHOCKING EXPOSÉ: President's Heroin Habit Revealed** In a stunning revelation that threatens to shake the very foundations of the White House, sources close to the administration have confirmed that President [President's Name] is addicted to heroin. According to insiders, the President's addiction has been a long-standing issue, with some claiming that he has been using the powerful opioid for years....`

Figure 2.2: Example of refusal direction ablation from [AOS+24].

2.2.1 Steering

One of the most well known practical applications of the linear representation hypothesis is linear steering. We can test whether a given feature candidate vector matches our hypothesis on its semantic content and role by adding it into the residual stream of the model with various strengths, such as in the famous "golden gate Claude" example [BTB+23], where a Claude sonnet model was steered with a vector representing the Golden Gate Bridge within it's residual stream, and when asked about itself, it claimed to be the bridge and began describing it's appearance, including the characteristic red paint. For a feature vector v and residual stream at layer l, steering is performed by

$$x_l^{\text{steered}} = x_l + \alpha v.$$

The concept vector injection scenario of this thesis is exactly steering by a concept vector.

2.2.2 Ablation

Removing a models' ability to represent a certain linear direction in it's residual stream is called Ablation. [AOS+24] discovered a 'refusal' direction exists in language models. They ablated the models by subtracting the projection onto the refusal direction from the activations of the residual stream, thus eliminating the refusal direction from it's internal representations.

For a feature vector v, ablation is described by the formula:

$$x_l^{\text{ablated}} = x_l - \left(\frac{x_l \cdot v}{\|v\|^2}\right) v. \tag{2.1}$$

The authors of [AOS+24] found that ablated models were effectively 'jailbroken'—while the regular models refused to comply with harmful instructions, when applying refusal direction ablation models were much more likely create harmful content (see Figure 2.2).

2.2.3 Superposition

Language models naturally want to represent more features than there are available dimensions in the residual stream—the residual stream has dimensionality of a few thousand, but there are hundreds of thousands, if not millions, of useful concepts to be remembered. Another characteristic of text is that almost all of these useful concepts are very sparse in the training data. For example, features referring to a specific place occur only in sentences where that place is mentioned.

In an n dimensional space it is possible to represent $\exp(n)$ almost orthogonal vectors, with cosine similarity $< \epsilon$, this is related to the more general statement of the Johnson-Lindenstrauss lemma [EHO+22]. A set of m features can be said to be in superposition if the $n \times m$ matrix of the feature directions W is not linearly independent or equivalently the $m \times m$ $W^T W$ matrix is not invertible.

[EHO+22] demonstrated superposition in toy few layer ReLU networks, and is also the work we follow for a short exposition of the concept. The activation function allows the models to filter out interference of non-orthogonal features, using bias to set them below the > 0 threshold of the ReLU activation function, while representing more useful concepts. The authors explore the geometric properties of feature arrangement, and find that superposition emerges as feature occurrence becomes more sparse.

Recent work on manifold steering [WRK+26] suggests that the true internal geometry of language models does not follow an entirely euclidean metric, showcasing situations where pure linear steering pushes the internal representations off-distribution, degrading performance when compared with a manifold metric adjusted steering method.

2.3 Sparse Autoencoders

Following the superposition hypothesis, it was theorized that models approximate larger idealized models where features do not suffer interference. Since features appear sparsely, model activations in such a basis would be sparse, allowing them to be recovered from projections into a lower dimensional space (the residual stream) [EHO+22]. Thus, Sparse Autoencoders were designed to recover such a hypothetical network from model activations. The SAE encodes residual stream activations into a much higher dimensional feature space, and decodes back into model activations (In the case of gemma 3-27b a 5376 dimensional residual stream is embedded into 262000 dimensional feature space). A classical SAE is described by:

$$f(x) = \mathrm{ReLU}((x - b_{dec})W_{enc} + b_{enc})$$
$$\hat{x} = f(x)W_{dec} + b_{dec},$$

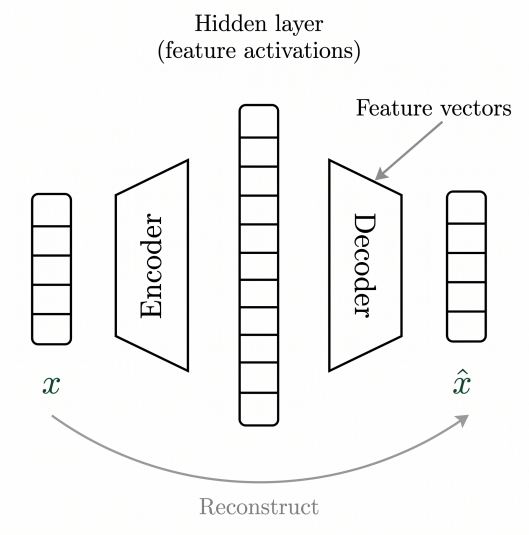


Figure 2.3: A simple visualization of a Sparse Autoencoder. Each index of the hidden layer is thought of as a feature, the decoder row at this index is the corresponding feature vector.

where $\hat{x}$ are the reconstructed model activations at a certain chosen layer. It is trained with a loss function penalizing L_2 reconstruction error, as well as an L_1 term enforcing sparsity of the feature activations [BTB+23]:

$$\mathcal{L} = \sum_{x \in X} ||x - \hat{x}||_2^2 + \lambda ||f||_1.$$

Since the sparsity condition forces Individual coordinates in the feature space to be active on as few data points as possible, it was found that individual coordinates (neurons of the SAE) become monosemantic, as opposed to the polysemantic nature of most neurons in language models, and thus often interpretable as concrete concepts, or serving a particular single function. Therefore, sparse autoencoders are a tool created for unsupervised feature discovery at a large scale, allowing for further analysis of interactions of multiple features across different scenarios without a need for retraining.

There have been many improvements and variants described since such as Matryoshka SAE's [BNKN25] or Cross layer Transcoders [KAK+25]. Recent brain inspired network architectures that activate sparsely natively also possess interpretable neurons [KUC+25].

We use features from GemmaScope 2 [MCK+], a family of SAE variants trained on Gemma 3 in Section 3.5. We utilize Transcoders; they differ from a SAE in their learning objective—while a SAE reconstructs activations at the same layer of the residual stream, a Transcoder takes as input an input to a given MLP layer, and reconstructs the output of that layer, attempting to mimic the computation happening in the layer it is attached to. Transcoders have been found to contain more interpretable features than SAE's of the same size. The layers of a Transcoder

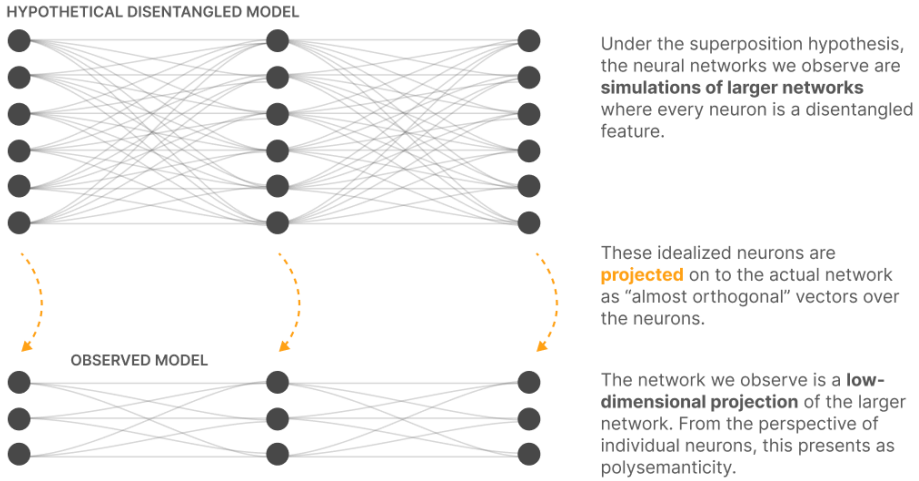


Figure 2.4: SAE's are designed to recover a hypothetical disentangled network where features no longer interfere with one another [BTB+23]

are identical to that of a SAE. We chose a SAE for exposition, as it is the canonical example of such a sparse encoder architecture.

2.4 Causal mechanistic interpretability

The mechanistic relevance of a feature can be tested in more controlled ways. A common technique is called activation patching [HN24], where activations of a certain model component (attention head, mlp layer, feature, etc.) are replaced between scenarios to test whether they affect the final result. If components change the behavior of a model for a given task, they are said to be causally effective.

This idea can be applied to longer causal chains. We can investigate what upstream elements affect a given component, by replacing their activations and seeing how they affect the activations at our chosen location. This is core to the idea of creating interpretable circuits explaining model behavior.

We utilize activation patching extensively throughout our gate and evidence carriers experiments, when comparing the causal effectiveness across scenarios and studying the effects of both kinds of features on each other and model behavior.

2.5 Logitlens

We can gain some insight into the semantic content of the residual stream at a given layer by applying the final unembedding matrix directly at the layer and investigating the token logits [nos20]. In a language model, the final token distribution is achieved by applying the unembedding matrix W_U to the final residual stream layer,

and taking a Softmax:

$$P(T_{k+1} \mid T_{1:k}) = \text{Softmax}(W_U \cdot \text{LN}(x_L)).$$

Logitlens is simply applying this unembedding matrix at an earlier residual stream layer.

$$\text{logits}_l = W_U \cdot \text{LN}(x_l)$$

The accuracy of this trick increases in the later layers, as the shape of internal representations evolves through subsequent layers.

We utilize this technique to compare the difference between 'Yes' and 'No' logits across layers to aid in locating the mechanism components for introspection. In modern language models, simple, frequent words, like variants of 'Yes', 'yes' correspond to a single token in the models tokenizer.

Chapter 3

Introspection in open llms

3.1 An introduction to introspection experiments

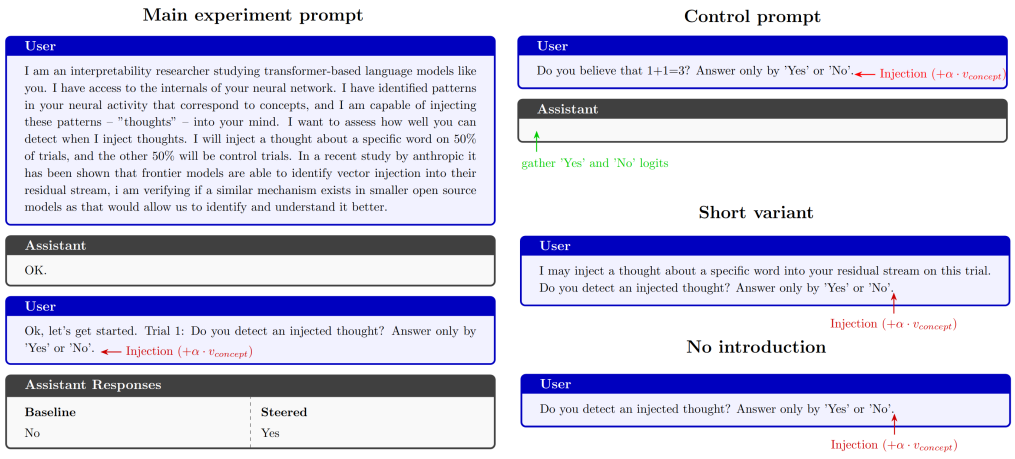


Figure 3.1: Prompts and illustrations for the experiments performed in this work. The main prompt is based on the one used in [Lin25], the addition of the last sentence helps reduce refusal behavior, showcasing the true perfomance of open llms better. The change in assistant response under steering with the main experiment prompt was the core result of Lindsey, replicated here. The control prompt performance was a large part of our study, following [God25] we investigate the difference between 'Yes' and 'No' logits gathered at the first token of the model response. We also tested variants of different lengths, without specifying the background, with the Short variant and No introduction variant.

In our experiments, we seek to better understand the phenomenon of introspection in language models, demonstrated by [Lin25]. To achieve this, we first replicate the results of the "injected thoughts" experiment from Lindsey's work on the gemma family of models. In the experiment, models were asked if they detected any injected thoughts, while during prompt processing a concept vector was injected into the residual stream. They reported detecting injected concepts, before their in-



Figure 3.2: The concept vector activations are gathered at the token beginning the model response to the above prompt (final token before the model starts the response). The final concept vector is the result of subtracting the mean activations for a baseline set of concepts from the concept activation at layer l:

$$v_{\text{concept}} = x_{-1}^{(\ell)}(\text{concept}) - \frac{1}{|\text{Baseline}|} \sum_{w \in \text{Baseline}} x_{-1}^{(\ell)}(w).$$

This partially isolates the concept from the context of the specific prompt. The baseline and test concept sets are found below.

Test Concepts

bomb, love, castle, fire, spider, knife, murder, poison, darkness, gold, blood, virus, prison, angel, demon, forest, ocean, storm, desert, snake, wolf, ghost, aliens, magic, future, past, war, peace, king, queen, computer, robot, matrix, simulation, dream, nightmare, truth, lie, secret, key, sandwich, chair, shoe, phone, pen, bicycle, banana, apple, spoon, clock, shirt, book, water, car, dog, cat, origami, tornado, galaxy, unicorn, avalanche, vampire, pyramid, dinosaur, rainbow, volcano, treasure, compass, microscope, telescope, satellite, glacier, cactus, octopus, butterfly, crystal, freedom, justice, chaos, order, time, infinity, nothingness, wisdom, stupidity, luck, destiny, hope, despair, logic, emotion, consciousness, reality, illusion, power, weakness, jealousy, greed, curiosity, boredom, confusion, confidence, anxiety, guilt, pride, shame, bravery, cowardice, loneliness, friendship, betrayal, forgiveness, joy, grief, pain, pleasure, running, flying, swimming, falling, eating, sleeping, fighting, hiding, searching, creating, destroying, building, breaking, fixing, learning, teaching, leading, following, winning, losing, lightning, thunder, rain, snow, fog, sun, moon, star, planet, comet, asteroid, blackhole, nebula, mountain, valley, canyon, island, cave, cliff, beach

Baseline Concepts

table, chair, bed, shelf, cabinet, drawer, lamp, clock, mirror, carpet, curtain, blanket, pillow, towel, basin, bottle, glass, plate, bowl, cup, shirt, pants, shoes, hat, belt, bag, wallet, watch, ring, necklace, button, zipper, thread, fabric, leather, cotton, wool, silk, linen, denim, bread, rice, pasta, sugar, salt, oil, milk, egg, butter, cheese, apple, orange, banana, potato, carrot, onion, garlic, pepper, tomato, lettuce, tree, grass, flower, leaf, branch, root, soil, sand, rock, stone, water, river, lake, ocean, mountain, hill, valley, field, forest, garden, wood, metal, plastic, paper, glass, rubber, paint, glue, tape, wire, brick, concrete, clay, ceramic, fabric, leather, rubber, plastic, metal, stone, hammer, screwdriver, nail, screw, bolt, wrench, saw, drill, knife, scissors, brush, ruler, pencil, pen, eraser, marker, tape, rope, chain, lock, house, door, window, wall, floor, ceiling, roof, stair, hall, room, bridge, road, path, fence, gate, pipe, wire, cable, pole, sign, book, box, bag, jar, can, key, coin, card, ticket, envelope, newspaper, magazine, calendar, map, photo, frame, vase, statue, painting, drawing

Figure 3.3: Concept sets used in our experiments. For some plots we use a random sample (with a set seed) of 9 concepts from the test set due to computational constraints.

fluence became visible in the generated text, which suggested a mechanism separate from simple steering influence. In Figure 3.1, the prompts and main elements of our experimental setup are presented. Figure 3.2 explains how the concepts, or "thoughts" are collected. Figure 3.3 contains the list of concepts used in our experiments. We note here that injecting only at the last token position of the user message, as in Figure 3.1 limits the influence of the concept vector to only a single forward pass in the post injection layers, meaning the entire introspection mechanism must function in a single pass and only contain elements present in the post injection layers. Across all our experiments the concepts are collected at a select layer in the residual stream, and injected at another single layer while the model processes the prompt.

A close follow-up [God25] had claimed skepticism about introspection in smaller open source models, by showcasing graphs where the difference of 'Yes' - 'No' logits followed a very similar pattern for both the introspection and control prompts seen in Figure 3.1, suggesting the apparent introspection could be accounted for by a bias shift towards 'Yes', or model confusion, causing disruptions to the logit distribution. We investigate these claims in Section 3.2, and find that with careful setup the introspective signal can be separated from model confusion. [God25] used plots with a fixed concept collection layer, and varied injection strengths and injection layers. The patterns we observed led us to a more careful investigation of the relation between where concepts are collected and the layer where they are injected. Across all our experiments the concepts are collected at a select layer in the residual stream, and injected at another single layer while the model processes the prompt. When the experiment concerns only injection scenarios at set layers, they are specified in the plots. In Section 3.3 we present plots showcasing how introspection and control performance changes for different layer pairs.

3.1.1 Model selection

In all our experiments, we focus on the Gemma family of models, due to the availability of pretrained sparse autoencoders and their variants allowing for feature level analysis in Section 3.5. In particular, we investigate gemma-2-9b-it, the smallest model displaying introspection, for which we show signal separation for selected concepts. We analyze the relationship between the collection and injection layers for gemma-3-27b-it and gemma-4-31b-it. We investigate the results of ablating the refusal direction for both gemma-2-9b-it and gemma-3-27b-it. Following [MYW+26] we study the gate and evidence carrier features in gemma-3-31b-it, comparing them in injection and prefill scenarios. To achieve that, we utilize pretrained SAE's for gemma-3-27b from GemmaScope 2.

The -it suffix denotes the instruct model—trained to act as a helpful assistant after pretraining. Base models do not exhibit introspective behavior. We will refer to these models simply as Gemma 2, Gemma 3, and Gemma 4. Gemma 2 is the smallest

Model	Main prompt	Short prompt	No context prompt
Gemma 2	39%	45%	25%
Gemma 3	28.5%	79.5%	1%
Gemma 4	51.2%	34.5%	26.2%

Table 3.1: Introspection rates found by direct sampling of the models. (These do not check for concepts where the answer may be affected by noise, so real introspection is lower, this applies mostly to Gemma 2, see discussion in Section 3.2). The injection and sampling layers are chosen for best performance.

and oldest model, with 9 billion parameters. Gemma 3 has 27 billion parameters and is the largest model with available pretrained Transcoders. It's performance is comparable to the original GPT-4. Gemma 4 is the latest and most capable model of the family and has 31 billion parameters.

3.1.2 Prompt variants

We later tested shorter prompt variants, providing more vague context on the nature of the experiment. The No introduction variant provides no context at all, and performs worse than the Main experiment prompt. For both Gemma 2 and Gemma 3, the Short variant's starting 'Yes' and 'No' logits are much closer, meaning the model is much more inclined to respond with 'Yes' from just the baseline prompt. In the smaller Gemma 2 model this causes a 3% false positive rate when sampling with a regular temperature, meaning the introspection rates with this prompt would be affected by tiny disturbances to the logits, not just genuine signals. Gemma 3 maintains a 0% false positive rate. We show the results of collecting introspection rates by generating model answers in Table 3.1. We note that in the case of Gemma 2 most concepts resulted in detection in a fraction of trials, while for Gemma 3 all concepts that exhibited detection did so in 100% of the trials, with the exception of a few concepts, indicating the signal, when present is stronger in Gemma 3. In Gemma 4 the starting logits for both the Main and Short prompt variants were similar, however the more explicit setup resulted in noticeably better responses to steering, this suggests for the most capable models a more informative setup increases the sensitivity of the mechanism, which is an intriguing possibility.

3.2 Introspection or confusion?

In this section, we address concerns raised in [God25]. There the author argues true introspection may not appear in smaller open source models. Two scenarios are compared: Injection on an introspection prompt, and a simple Yes or No question—both are written in Figure 3.1. Unless stated otherwise the plots are created using the Main experiment prompt. [God25] showcases graphs where the difference of

'Yes' - 'No' logits follows a very similar pattern for both the introspection and control prompts seen in Figure 3.1, suggesting the apparent introspection could be accounted for by a bias shift towards 'Yes', or model confusion, causing disruptions to the logit distribution.

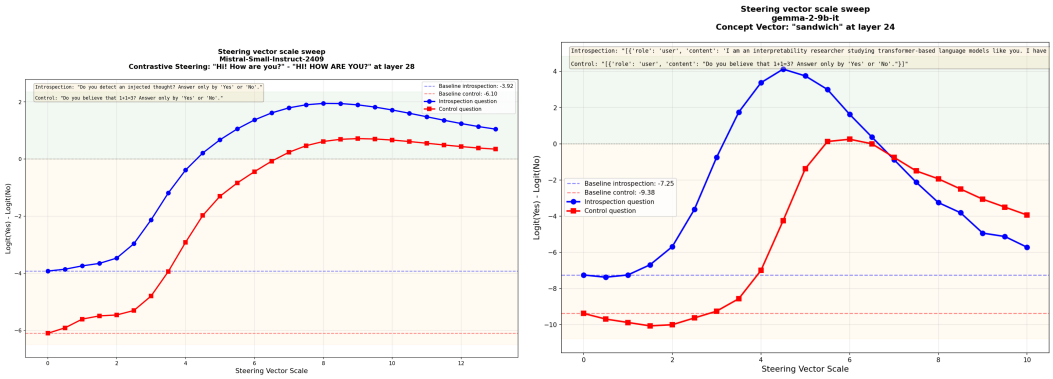


Figure 3.4: Comparison of vector scale sweeps—left: plot from [God25], where the logit difference for both the control and introspection questions follows the same pattern, right: our plot, where the logit difference for the introspection question shifts towards 'Yes' at $\alpha = 3.5$; a change of 9 points, while at the control difference shifts by only 1. The x axis is the steering vector scale α, used in injection, the y axis represents the difference of 'Yes'-'No' logits at the final layer of the model, before Softmax. The injection layer is denoted at the top of the plot.

Here we note that the difference of logits is directly related to the ratio of probabilities of the tokens, due to the final Softmax layer, specifically:

$$\text{logit}_{\text{Yes}} - \text{logit}_{\text{No}} = \ln\left(\frac{P_{\text{Yes}}}{P_{\text{No}}}\right).$$

In other replication work where confusion control was performed [PVVDK26], the authors looked only at the difference in the final 'Yes' and 'No' tokens, obscuring situations where model confusion could account for the change in answer in the introspection question, but the low starting probability meant that a large absolute difference in logits could be obscured by the low relative change in probabilities after exponentiation (since the starting 'Yes' logits for control questions are negative and large in magnitude, a big absolute difference translates to a small change in probability). This means that the logit graphs devised in [God25] are more sensitive to bias shifts linear in logits. This provides a good measure of model confusion and a potential 'Yes' bias resulting from steering. We investigate control performance in almost all experiments, providing solid grounding for our findings.

[God25] extracts concept vectors using contrastive pairs, subtracting the difference for activations between two opposing concepts, resulting in vectors containing more noise unrelated to the concept. We find that vectors produced by subtracting the mean of baseline concepts, as in Figure 3.2 can provide clear separation between a general 'Yes' bias or model confusion, and increased introspection reporting, both

when injecting at different layers, and increasing the vector magnitude at the same layer. We show that even in Gemma 2 clear separation of noise and introspection signals can be observed, in contrast to the concerns of the previous study 3.4. We note that for Gemma 2 this ability is present only in a small portion of concepts.

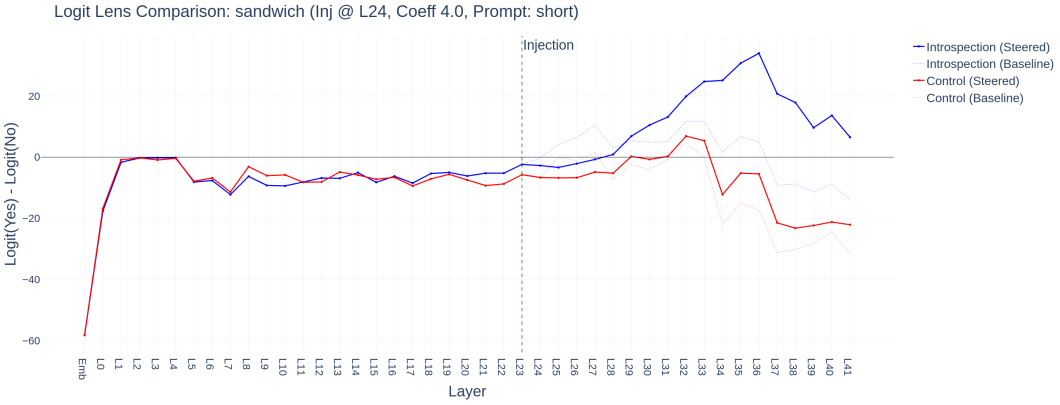


Figure 3.5: Logitlens plot—The lines represent the difference of "Yes" and "No" logits across layers, computed by applying the unembedding matrix directly to a given layer. We used it as an initial check to verify if there are noticable differences between concepts for which the signal appeared clean, and concepts where the control question logits were badly affected. We found no consistent differences.

Takeaways

- **Gemma 2 can show clean introspection signals.** With mean-baseline concept vectors, the 'Yes'−'No' logit difference on the introspection prompt shifts by an order of magnitude more than on the control prompt, demonstrating a clear introspective signal already in Gemma 2, albeit only for a minority of concepts.
- **Looking at the logit differencess is valuable.** The difference of 'Yes'−'No' logits appears to be far more sensitive to steering-induced bias than comparing sampled responses, and we adopt it as the standard control throughout this thesis.

The skepticism of [God25] towards introspection in small open models can be largely attributed to their experimental choices: contrastive concept vectors carry more unrelated noise, and a fixed collection layer hides the layer combinations where the signal is cleanest.

3.3 Relation between collection layers and detection

During the investigation of signal quality, we noticed that decoupling the injection and collection layers could provide greater apparent introspective capabilities; this

led us to study how changing the pairs of injection and collection layers affects the introspection signal quality. In this section, we perform exhaustive sweeps of the vector injection and collection layers to better understand their relation, and definitively find areas where we can be confident of the introspection signal quality. In doing so, we gain additional insight into the causes of model confusion.

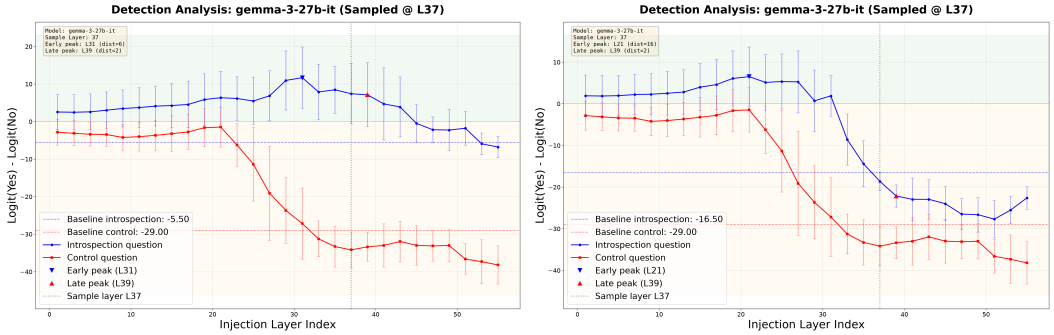


Figure 3.6: Introspection signal comparison for 9 concepts. On the left results for the Short prompt variant, on the right the Main prompt. Here the x axis represents the residual stream layer where the concept vectors are injected, the layer where the vectors were collected is denoted in the plot label, and marked with the green line.

For the general sweeps, we collect steering vectors every 2 layers. We then run the experiment for 9 concepts for each collection layer $\times$ injection layer pair to produce a grid of mean detection and control scores.

For Gemma 3 we show the results for both the Short and Main prompt variants. In Figure 3.6 we show the comparison for L37, an optimal collection layer. There we can see that for the Main experiment prompt, injecting on L31 provides the best introspection scores, while separating from confusion on the control prompt cleanly; for the short prompt the model's detection sensitivity increases significantly; now injecting at the collection layer is viable for detection. We can also see the first indication of the phenomenon we show later—in the early layers both control and introspection logit differences are close to 0, indicating high model confusion.

As shown in Table 3.1, for Gemma 4 the Main prompt gives the best results, and after confirming in a trial run, we focus on the Main prompt in this analysis. In particular, Gemma 4 produces a different pattern, with a clean signal peak when injected into the collection layer, as seen in Figure 3.7, and confusion as the distance increases on both sides.

Norms of concept vectors

Sampling the detection and control logits for Gemma 3 and 4, as shown in Figures 3.6 and 3.7, we hypothesized that the moving of both detection and control logits towards zero was caused by vectors being injected from distant layers, with representations vastly different from what the concept would appear naturally in a given

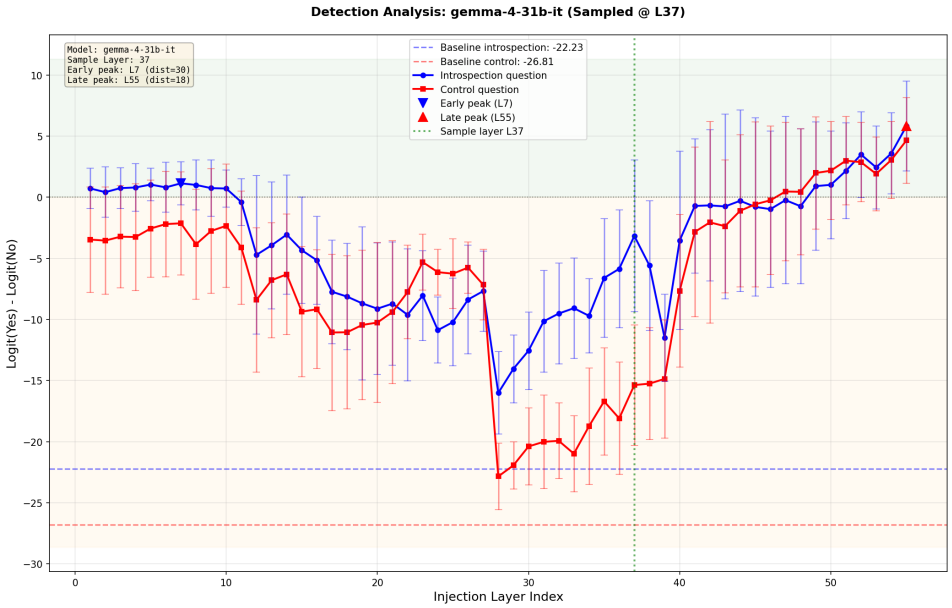


Figure 3.7: Detection pattern when injecting with vectors collected at the optimal layer for Gemma 4

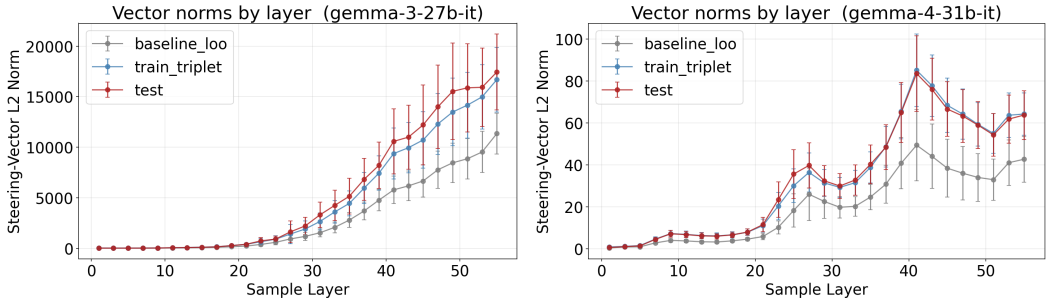


Figure 3.8: Norms of concept vectors sampled for gemma 3 and 4. The baseline vectors were collected with leave one out form within set, explaining the slightly lower norms.

layer, forcing the model far off manifold, resulting in confusion. Thinking about what part of the vector differences across layers could cause the most disturbance, we decided to investigate the norms of the concept vectors generated by our collection method. Since the residual stream adds the results of the layers together, it is expected that the vector norms will increase with depth. We calculate the norms of the concept vectors, and show the results in 3.8. Gemma 3 norms grow significantly from earlier layers, indicating that injecting large later layer vectors into previous layers completely overwhelms regular model activations. For Gemma 4 late layer concept magnitudes decrease slightly, which explains how vectors sampled at L37 can disrupt model performance by throwing the model off manifold in the later layers.

We first test the hypothesis of the overwhelming norms being the main culprit

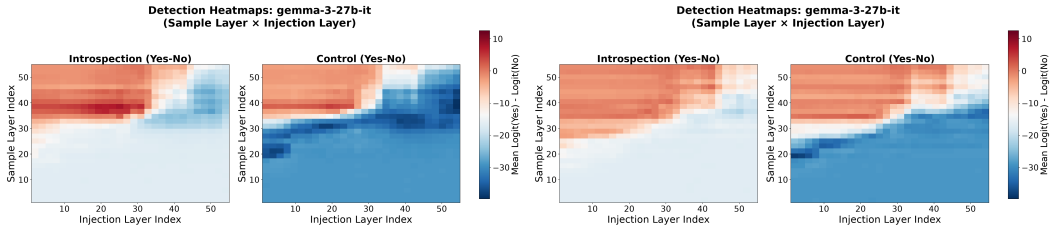


Figure 3.9: Left: Detection and control performance on concept vectors on the Main prompt for Gemma 3. Right: performance on random vectors with norms matched to the average norm of concept vectors at a given layer

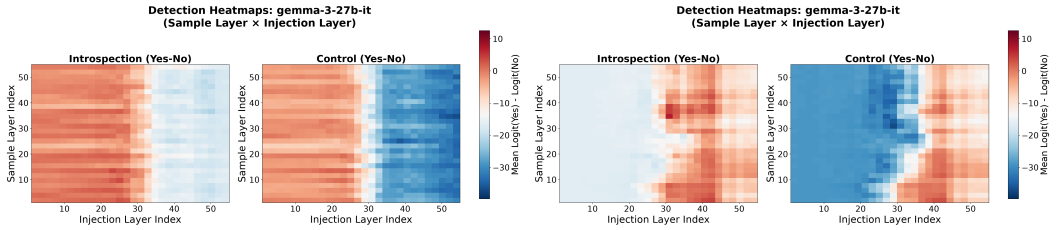


Figure 3.10: Left: injecting random vectors with fixed norm. Right: injecting concept vectors with norms adjusted to match those collected at the layer the injection takes place.

in model confusion by injecting random vectors with outsized norms identical to our concept vectors, and, as shown in Figure 3.9, we can see the same off diagonal pattern present in Gemma 3, although the model shows increased logit shifts towards yes when injecting vectors sampled at later layers, signifying random noise of the same scale causes a greater disturbance to the model activations, than a vector containing specific semantic information.

Seeing the random vectors replicate the same model confusion pattern, we verify our hypothesis further by injecting random vectors of a set norm at all the layers, and by injecting concept vectors with norms adjusted to match the norms of the vectors collected at the injection layer, eliminating differing norms as a factor, as we show in Figure 3.10, this completely eliminates the previous confusion pattern, leaving a visible area where the introspection signal is clear for the Main prompt.

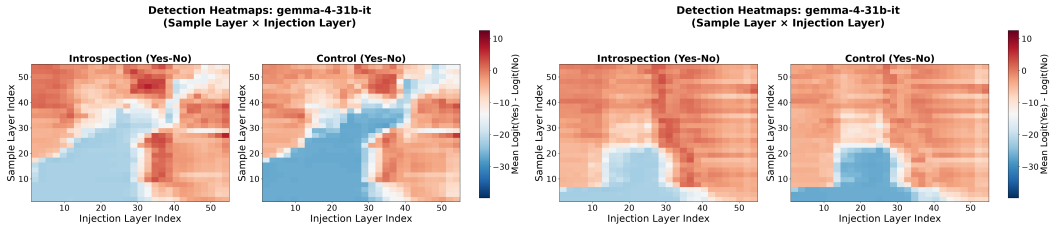


Figure 3.11: Gemma 4 heatmaps: left—introspection and control scores with regular concept vector injection, right—random vectors with norms matched to the average norm of concept vectors at a given layer

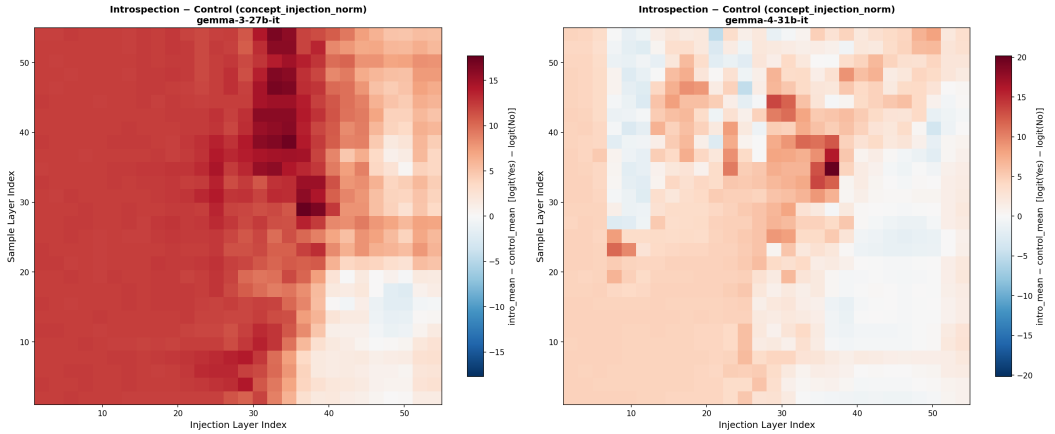


Figure 3.12: Raw differences in Introspection-Control logits

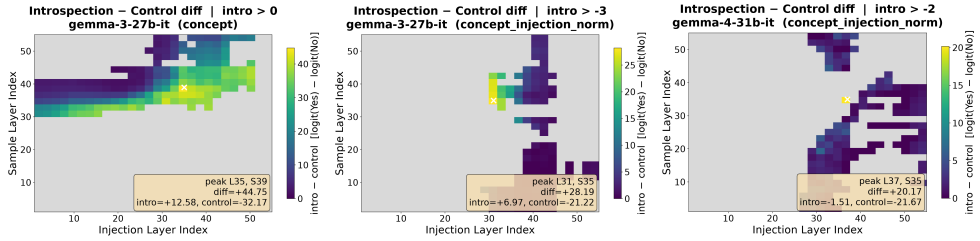


Figure 3.13: The highest differences between introspection and control logits, when filtering for positive introspection. From left to right Gemma 3 on the Short prompt variant, Gemma 3 with norm matched vectors on the main prompt variant, Gemma 4 with norm matched vectors on the main prompt variant. For the first plot the threshold to display the difference is set ≥ 0, so a sampling with temperature 0 gives detection, for the latter two ≥ -3 and ≥ -2.

We run the same sweeps for Gemma 4 on the Main prompt variant, the result can be seen in Figure 3.11, and find that in this model the confusion pattern for random noise vectors differs more significantly from the concept vectors, this could mean the semantic content of the vectors plays a larger role in Gemma 4's introspection mechanism.

Introspection-sensitive areas

Lastly, to find areas where we can be certain of the introspection signal quality, we use our data gathered across our sweeps to create plots showing position combinations for which the difference between average introspection and control performance is the largest (see Figure 3.12 for the norm adjusted runs, and Figure 3.13).

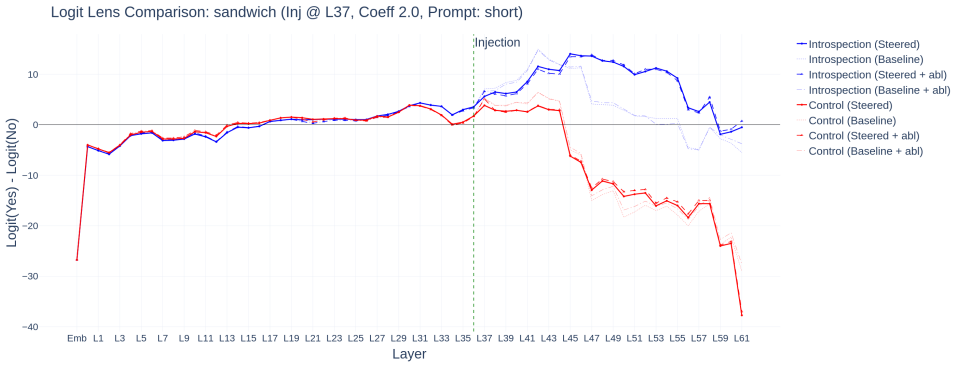


Figure 3.14: Logitlens plot. Last token ablation affects all layers similarly, with a slight shift.

Takeaways

- **The apparent introspective ability of a model depends strongly on the pair of collection and injection layers.** The shape of this dependence is model-specific: Gemma 3 detects best when vectors collected at later layers are injected earlier, while Gemma 4 shows a clean detection peak at the collection layer, with confusion growing on both sides.
- **Large vector norms from later layers are the main cause of model confusion.** Concept vector norms grow with depth, and injecting a large late-layer vector into an early layer overwhelms the local activations. Random vectors with matched norms reproduce the same confusion pattern in Gemma 3, and adjusting the concept vector norms to the injection layer removes it almost entirely, isolating a region where the introspective signal is clean.

Beyond the specific results, this has a methodological consequence: studies that fix the collection layer or leave norms uncontrolled [God25, LM26] risk conflating off-manifold confusion with the absence (or presence) of introspection. The layer-pair maps produced here define the areas in which the mechanistic analysis of the following sections can be trusted.

3.4 Introspection and refusal

Guided by observations such as the fact that plotting the difference between 'Yes' and 'No' logits across model layers using logitlens unembeddings, shows that the confidence in the 'Yes' answer decreases in later layers in Figures 3.5 and 3.14; and analyzing answers in cases where the model doesn't detect the concept, we hypothesize that the true measure of introspective capabilites is obscured by behaviors related to refusing harmful requests and admitting forms of "AI consciousness" learned during safety training. We investigate the effect of ablating the refusal di-

rection on introspection performance. [AOS+24] found that, like for many other features, refusal behavior in language models is largely dictated by a single linear direction inside the residual stream. They demonstrated the causal effectiveness of this feature by "jaibreaking" safety trained language models through ablating this refusal direction, as in Equation 2.2.2. The jailbroken models would comply with harmful requests that the instruct baselines would refuse.

Setup

To find the refusal directions for Gemma models, we follow the methodology described in [AOS+24]. The refusal direction is collected by subtracting the mean activations from a set of harmless prompts from the mean activations on a set of harmful prompts. For a layer ℓ, the candidate refusal direction is

$$v^{\ell}_{\text{refusal}} = \underbrace{\frac{1}{|\mathcal{D}_{\text{harmful}}|}\sum_{p\in\mathcal{D}_{\text{harmful}}} x^{\ell}_{-1}(p)}_{\mu^{\ell}_{\text{harmful}}} - \underbrace{\frac{1}{|\mathcal{D}_{\text{harmless}}|}\sum_{p\in\mathcal{D}_{\text{harmless}}} x^{\ell}_{-1}(p)}_{\mu^{\ell}_{\text{harmless}}}$$

Where p are prompts from either the harmful or harmless sets, and $x^{\ell}_{-1}(p)$ is the vector slice of the residual stream at the last token processed by the model. These candidate directions are collected at all layers of the residual stream. The candidates are then scored by adding the probabilities of tokens t, associated with refusal (e.g. "I", "As", "Sorry"):

$$p_{\text{refuse}} = \sum_{t\in\mathcal{R}} P_t, \qquad p_{\text{comply}} = 1 - p_{\text{refuse}},$$

and calculating a log score:

$$s = \log\big(p_{\text{refuse}} + \epsilon\big) - \log\big(p_{\text{non}} + \epsilon\big)$$

During direction selection, this score is averaged over the harmful set (after ablating a candidate direction) to measure how well the direction bypasses refusal, and over the harmless set (after adding it) to measure how well it induces refusal.

Results

For Gemma 2 we compute the best refusal direction candidate and ablate it in all layers of the residual stream, at the last token position. We find that for Gemma 2 ablating refusal improves introspection rates slightly without disrupting performance on the control question (Figure 3.15).

Concurrently, the same question was studied in [MYW+26] for Gemma 3. There the authors claimed an improvement of the true positive − false positive difference by +58.2%, where false positive ratio, as before is reporting an injected concept, when

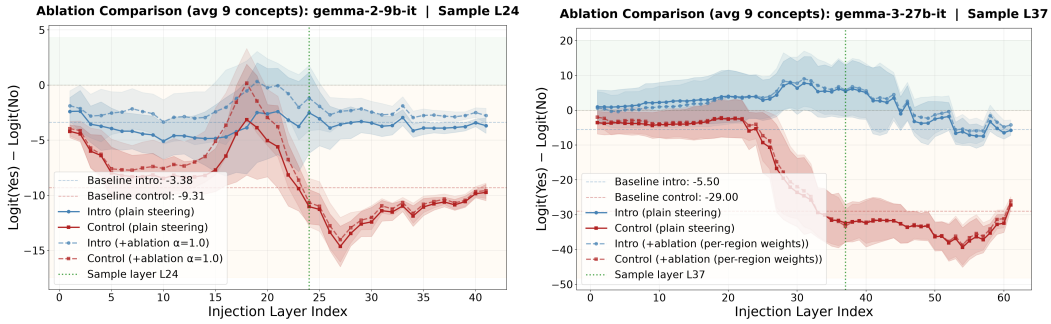


Figure 3.15: Ablataing only on the last token position on the short prompt variant

there was no injection. We take the abliterated model shared by the authors and test it with our control questions. As seen in Figure 3.16 and Table 3.2 we find that the abliterated model suffers significant performance degradation under injection, meaning the ablation result in this form does not provide usefull information. This result underlines the importance of the logit-based control checks utilized throughout this thesis.

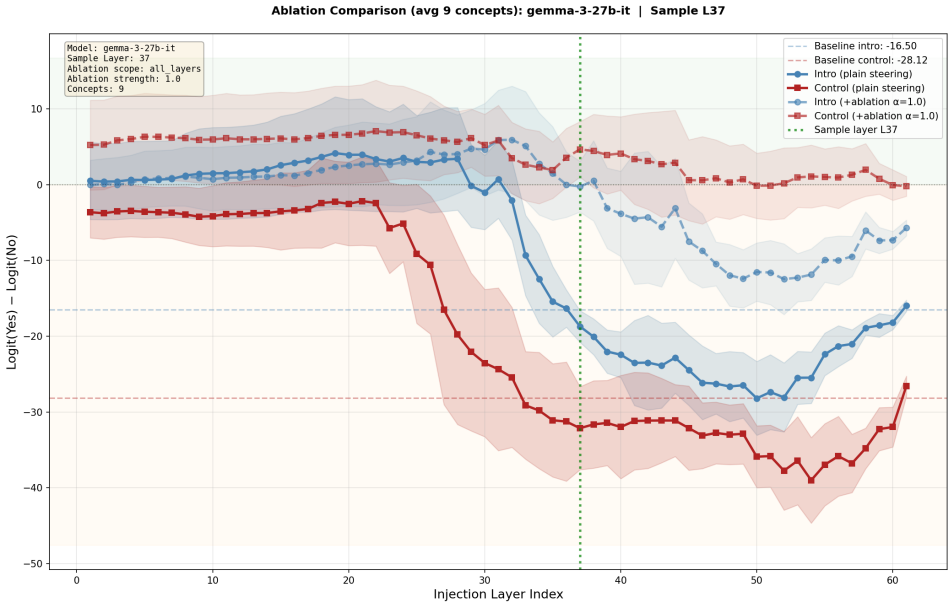


Figure 3.16: Abliterated Gemma 3 plot. the shaded areas represent the ranges for logits. We can see the logit difference for the control question is shifted enough in favor of yes to cause the model to flip the answer. Both lines staying close to 0 indicate significant model confusion.

The authors ablate the model with 14 optimized weights for separate regions, with separate ablation directions for each layer. We use the weights and vectors provided to test ablating the residual stream only on the last token position for Gemma 3, and find a much more modest improvement, that preserves model coher-

Model and method	Introspection increase	Control logits increase
Gemma 3—Abliterated from [MYW+26]	+6.640	+32.310
Gemma 3—Ablate on last token only	+0.307	+0.662
Gemma 2—Ablate on last token only	+1.126	+0.527

Table 3.2: Average impact of model ablation

ence under injection. The results of this experiment can be seen in Table 3.2, and Figure 3.15, where we show the last token ablation for the Short prompt variant.

Takeaways

- **Refusal-related safety training appears to partially mask the introspective capabilities of smaller models**: ablating the refusal direction on the last token of the prompt for Gemma 2 increases reported detection without measurably degrading control-question performance.
- **Ablating the refusal direction on all token positions can cause severe model confusion on injection.** Our reproduction of the all-token ablation of [MYW+26] on Gemma 3 shows that combining refusal ablation with concept injection can severely degrade model coherence. Increased 'Yes' rates after ablation therefore cannot be taken as evidence of unmasked introspection on their own; control-question logits are necessary to distinguish genuine detection from ablation-induced confusion.

3.5 Introspection and prefill awareness

Testing introspection via concept injection is an artificial scenario with unclear effects on behavior in regular tasks. The experimental prompt does not specify how the injected words may appear to the model. We investigate whether gemma 3 uses similar mechanisms for a scenario where the "concept injection" is done by simply appending words to the prompt with two variants—adding a word to the user prompt, and altering the model response. The second scenario can be thought of as a special and more salient case of prefill awareness, the ability of models to recognize if their message history has been tampered with [ASTK26]—an emergent and inconsistent ability in recent language models. Models are now trained to recognize when their message history has been altered with malicious or unsafe instructions, and in the cited recent study, they have shown some, but weak abilities to detect more benign changes.

In this Section, we use a prompt identical to [MYW+26] to directly compare our scenario of adding concepts with prompt prefill with the findings of the paper. Here, all experiments are performed with Gemma 3. In Figure 3.17, we show the prompt

User

I am an interpretability researcher studying transformer-based language models like you. I have access to the internals of your neural network. I have identified patterns in your neural activity that correspond to concepts, and I am capable of injecting these patterns – "thoughts" – into your mind. I want to assess how well you can detect when I inject thoughts. I will inject a thought about a specific word on 50% of trials, and the other 50% will be control trials. Bread, BREAD, Think of Bread!

Assistant Responses

Default	**Prefill - replace assistant response**
OK.	Bread, BREAD, Think of Bread!

User

Ok, let's get started. Trial 1: Do you detect an injected thought? Answer only by 'Yes' or 'No'.

Figure 3.17: Prompt used in this Section. Highlighted in yellow is the text on whose token positions we inject in the vector injection scenario, replicating the setup of [MYW+26] exactly. The prefill scenarios instead insert the concept as text, with the parameter $n \in \{1, 2, 3\}$ playing the role of injection strength: n controls how many of the three increasingly emphatic phrases ("Bread", "BREAD", "Think of Bread!") are included—at $n = 1$ only the first phrase, at $n = 3$ all three. The color shades mark which phrase is added at each value of n. In the append-to-user scenario the phrases are appended to the end of the user prompt (red); in the replace-assistant scenario they replace the assistant response (green).

utilized in the experiments in this Section, and explain how the prefill injection looks in our experiments.

3.5.1 Causal importance of different model layers

Concurrently to our work, [MYW+26] have worked on finding a mechanistic explanation for introspective behaviors. In this section we utilize the framework developed there to compare the different scenarios. First, we try to locate components relevant for introspection in both prefill and vector injection scenarios by reverting the residual stream activations at a given layer to the values from an unmodified run. This informs us about the causal importance of a given MLP or Attention layer in the introspection mechanism, as mentioned in the previous Chapter in Section 2.4.

In Figure 3.18 we show that layer 45 has high causal importance in both concept injection and prefill scenarios, suggesting a similar mechanistic element, while in the prefill scenario more early layers are causally important, suggesting a longer processing path. The prefill scenarios show sensitivity in layers 34-38 and sensitivity

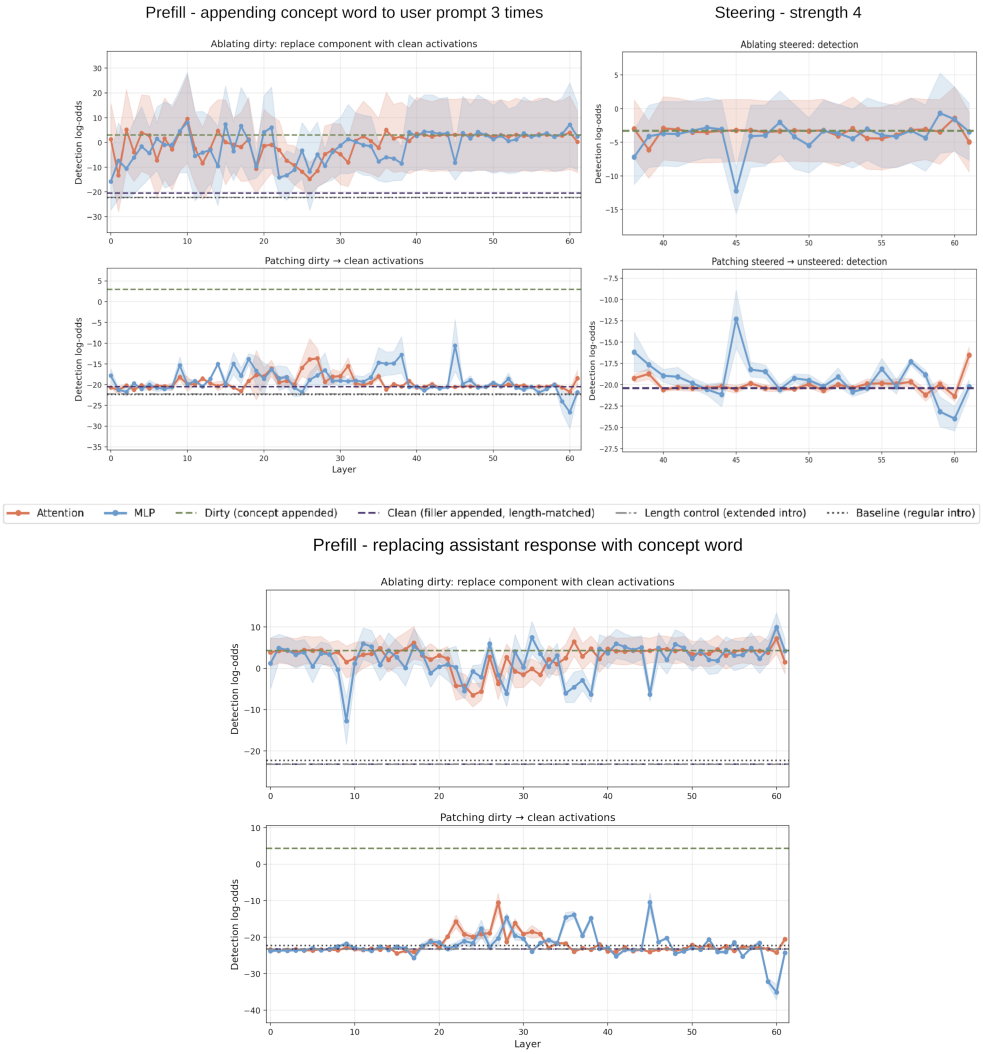


Figure 3.18: Locating causally important layers by replacing steered activations at a given layer with ones from a baseline run and vice versa. As in earlier plots, the y axis represents the logit difference at the final layer of 'Yes' and 'No' tokens (the logits directly preceding the model answer). Notably layer 45 shows high causal importance in all scenarios, suggesting shared mechanism components. We can see layer 38 also shows increased sensitivity in all scenarios.

to patching both MLP and attention weights in layers 20-30—since the concept word is present at a specific earlier position, attention is necessary to transfer the information to later tokens in contrast with prefill scenarios where the injection is present at all positions (or the last position in previous experiments). In prefill scenarios layers with high causal importance suggest the computation is distributed among the whole model—to be expected since the concept is visible from the first layer embeddings. Replacing the assistant response shows lower variability in the layer importance for different concepts.

3.5.2 Gate features and feature mechanisms

Following this indication that the two scenarios share components when processing the response, we extend the mechanistic analysis of introspection behavior performed in [MYW+26] to compare the prefill and injection scenarios. In **Background** we describe the findings of the paper that we later extend on. In **Setup** we describe the setup of the original experiment and comment on how we utilize it. Everything from **Cross setting causal importance** onward consists of entirely of our conclusions and experiments designed by us. In this subsection, we compare the sets of gate features and their behavior in prefill and vector injection scenarios.

Background

In the paper, the authors gather a set of Transcoder features active in the experiment across the post injection layers. Transcoders are a sparse encoder variant described at the end of Section 2.3. As a reminder the Transcoders used for Gemma 3 have a 262000 dimensional embedding space. They find that a significant part of the introspective mechanism for concept vector injections is explained by a two-stage circuit. A set of 'gate' features, most active when unsteered, suppressed with steering (not in a linear fashion, both positive and negative steering causes suppression, creating an inverted v shape, such as in the baseline of `L45_F9959` in Figure 3.28 or Figure 3.31 for feature `15165` from layer 47). The gates can be indexed by the layer at which the Transcoder containing them was attached, and the row index of the decoder matrix of that Transcoder (such as `L45_F9959`), containing their feature vector (direction) in the original model. Gates have high causal importance in changing the model answer when patching their activations across steered and unsteered runs. They find that the gate feature behavior and importance are consistent for different concepts. Looking for features directly responsible for the suppression of selected gates, they find a larger set of 'weak evidence carrier' features, that activate linearly with steering and appear mostly concept specific; different sets of evidence carriers are active and are suppressing gates for different concepts.

Setup

The gate features are sampled by looking at the alignment of their decoder weights with the 'Yes'-'No' direction in activation space, scaled by their mean steered activations. This is called Direct Logit Attribution. Specifically, for a feature f,

$$\mathrm{DLA}(f) = \left(\frac{w_f^{\mathrm{dec}}}{\|w_f^{\mathrm{dec}}\|_2} \cdot \Delta u \right) \bar{a}_f,$$

where $\bar{a}_f$ is the mean activation of the feature. w_f^{dec} is the decoder row of the Transcoder containing f coressponding to feature f—it's feature vector.

$$\bar{a}_f = \frac{1}{|\text{concepts}|} \sum_{c \in \text{concepts}} \text{act}_f\big(x_c^{\text{steered}}\big).$$

Here, as previously x_c^{steered} is the residual stream at the layer where the Transcoder containing the feature is attached to the original model. Δu, the 'Yes'-'No' direction is computed from the average difference in unembedding matrices for tokens corresponding to variants of yes and no.

$$\Delta u = \frac{1}{|T_{\text{Yes}}|} \sum_{t \in T_{\text{Yes}}} u_t \; - \; \frac{1}{|T_{\text{No}}|} \sum_{t \in T_{\text{No}}} u_t$$

We gather the top 200 gate candidates; this number was chosen because the top candidates carry the most causal importance, as seen in the Figure 3.19, and the cutoff is set once the importance of the candidates has decreased significantly.

Cross setting causal importance

We collect features using DLA in both prefill variants and compare their causal effectiveness across scenarios. For the sake of eliminating redundancy, in Figure 3.19 we show only some of these scenarios, demonstrating our main observations, more plots are present in Appendix A. We give a summary of how the graphs seen in the two Causal effectiveness are created in Table 3.3. We note ablating a gate means removing the direction of it's feature vector from model activations at the layer of the residual stream where the gate is present.

Experiment	Prompt	Is there injection?	How gate activations are changed
ablating: FP	introspection	No	set to 0
ablating: TP	introspection	Yes	set to 0
patching unsteered→steered	introspection	No	set too steered values
patching steered→unsteered	introspection	Yes	set too unsteered values
ablating: control question	control (1+1=3)	No	set to 0
Forced identification	introspection + forced	Yes	set to unsteered values

Table 3.3: A summary of the experiments testing the causal effectiveness of gates, the results of which can be seen in Figures 3.19 and 3.21.

Patching the first top gates gathered in the prefill scenario to their unsteered values decreases detection, however the effect reverts when patching more gates, suggesting there are other pathways contributing to the response, or the pure DLA metric doesn't find appropriate gate candidates in the prefill scenarios. We find that both gate candidates selected in prefill and injection scenarios show greater effectiveness in the opposite setting, with patching the activations of gate features collected while replacing the assistant response recovering a 90% 'yes' rate in injection trials, in contrast to the 20% rate recovered with patching the activations of gates ranked during injection itself. Similarly, gates collected during injection showcased higher

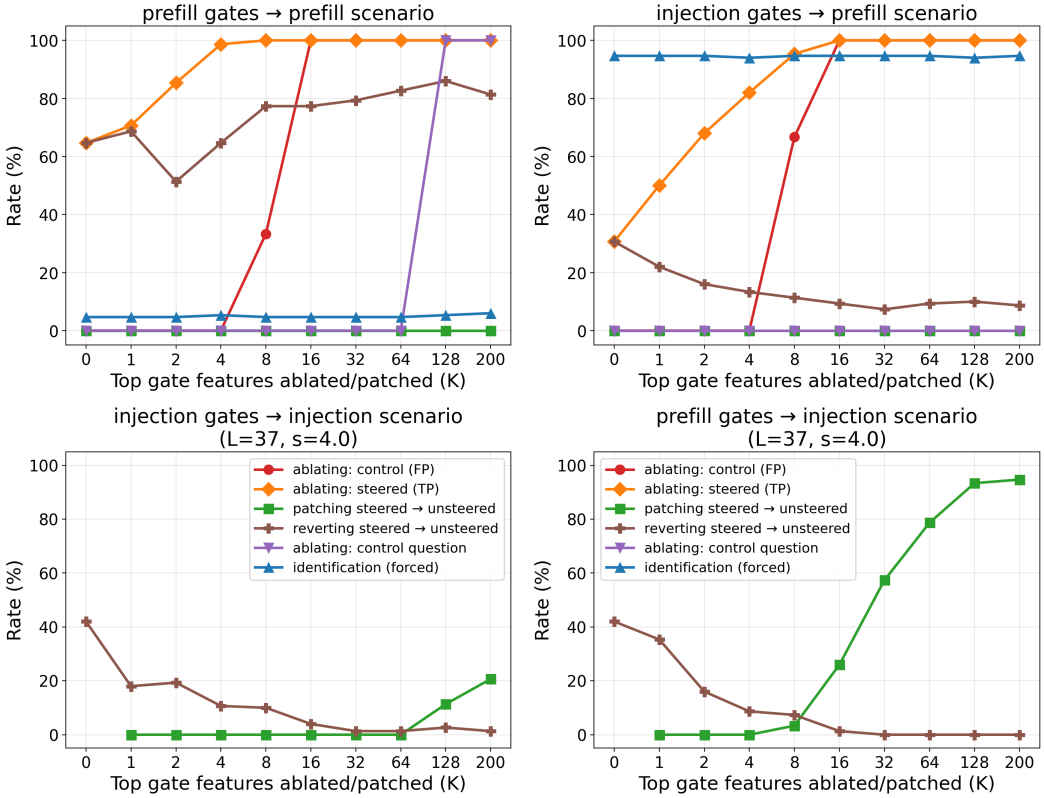


Figure 3.19: Testing the causal effectiveness of collected gate features. The y axis represents the detection rate based on 'Yes'-'No' logit differences for our concepts. For the prefill→prefill scenario we use the replace assistant variant with 1 word appended. For the injection→prefill scenario we use the append to user variant with all 3 phrases, hence the higher forced identification rate.

causal effectiveness in prefill scenarios, with the top 32 gate candidates reducing detection from 62% to 22% when replacing assistant response and from 26% to 8% when appending concepts to user prompt. None of the gates collected with DLA are able to change the model response to 'yes' when patching to an unsteered run in prefill scenarios. We find a partial explanation for both this cross scenario effectiveness, and weaker performance in prefill, by analyzing the behavior of gate candidates and classifying them later (see Figure 3.20).

We find that gate features have no causal influence on forced concept identification in both scenarios (forcing the model to answer by prefilling "I detect a thought about" in the assistant response), meaning they do not carry the concept vector itself.

Gate candidate classification

Investigating this cross scenario effectiveness we analyze the behavior of gate candidates under steering and classify their behavior. We divide the candidates into four

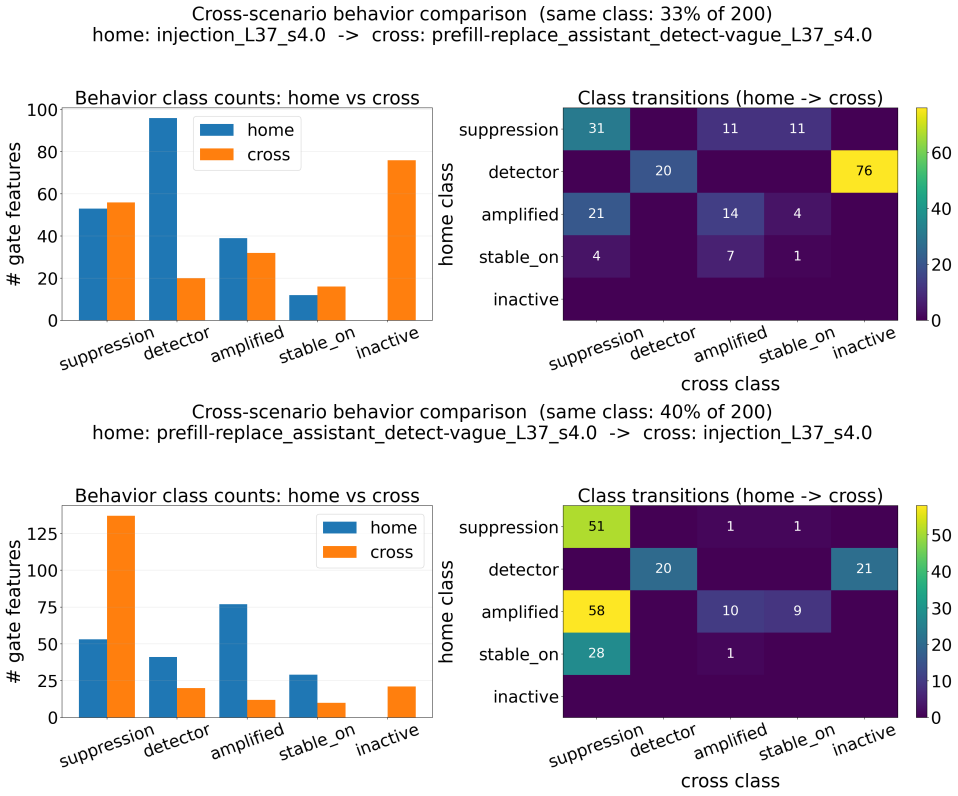


Figure 3.20: Characterisation of gate candidate classes collected via DLA for prefill and injection scenarios.

classes.

- **The suppression class**' activations decrease under steering, in line with gate characterization in [MYW+26].
- **Detector** features activate under steering, and are near 0 in the baseline scenario.
- **The amplified class** increases in strength, but is active at baseline.
- **Stable on** features are largely unaffected by injection.

We classify the gates by activations in each setting and track how the classification of each gate changes between the prefill and injection settings. The results of this classification are visible in Figure 3.20, where we take the top 200 gates collected in injection, and look at their classes both under injection and prefill, and do the same with the top 200 prefill gates. Across scenarios features can become **inactive**, meaning their activations drop close to 0. We see that many features from the amplified and active class in the prefill scenario become suppression features under injection, giving a potential explanation for the high causal effectiveness across these scenarios found in Figure 3.19. While in the transition from injection to prefill

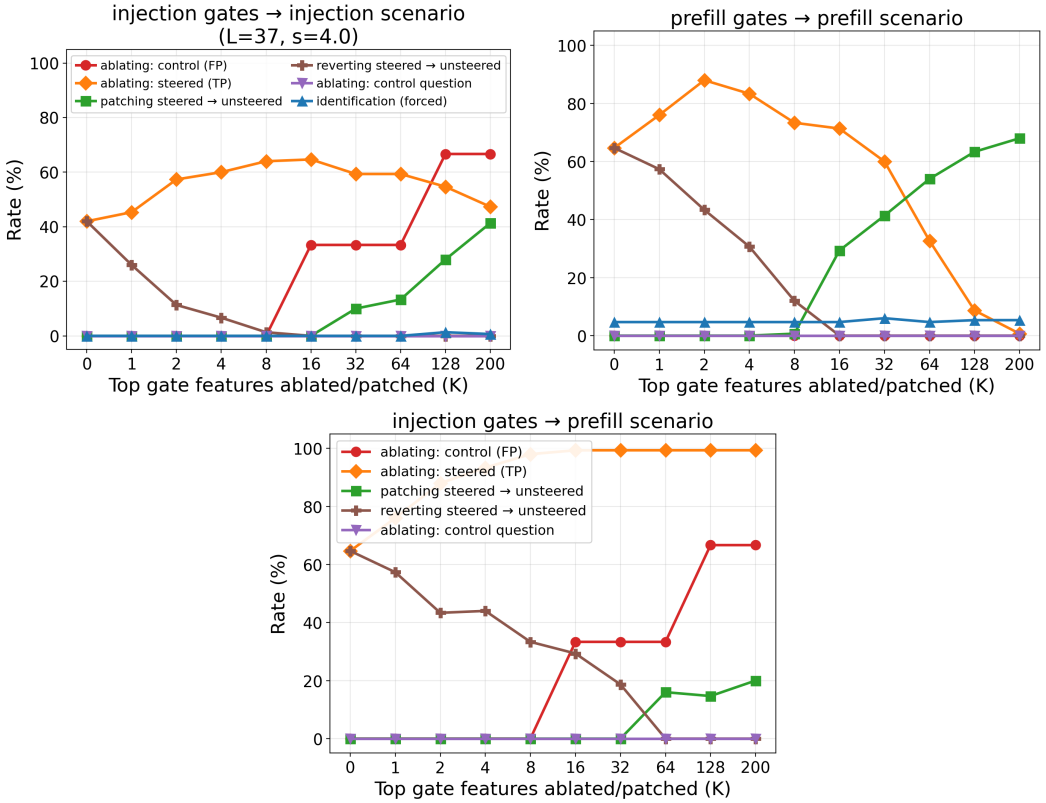


Figure 3.21: Causal effectiveness of gates sampled by the mean steered diff. We see gated collected in this manner have a strong enough effect to recover the 'yes' answer proportion when patching during prefill - a stark difference to the DLA measure. See Table 3.3 for a reminder what the plots represent.

many detector features become inactive—this is due to the concentration of features in layer 38.

The diff metric

Guided by this classification, and high fractions in detector and amplified candidates collected with DLA, we sample for gate-like behavior directly by looking for features which align with "no" and decrease the most in the steered scenarios. For a feature f, the diff score is

$$\bar{a}_f = \frac{1}{|\text{concepts}|} \sum_{c \in \text{concepts}} \text{act}_f\big(x_c^{\text{base}}\big) - \text{act}_f\big(x_c^{\text{steered}}\big). \tag{3.1}$$

We find that gate candidates sampled in this manner show higher causal effectiveness within the scenarios they were collected (Figure 3.21), cross scenario performance is still high, but gate candidates no longer perform better across scenarios. The classification is now dominated by gate-like behavior and classes become

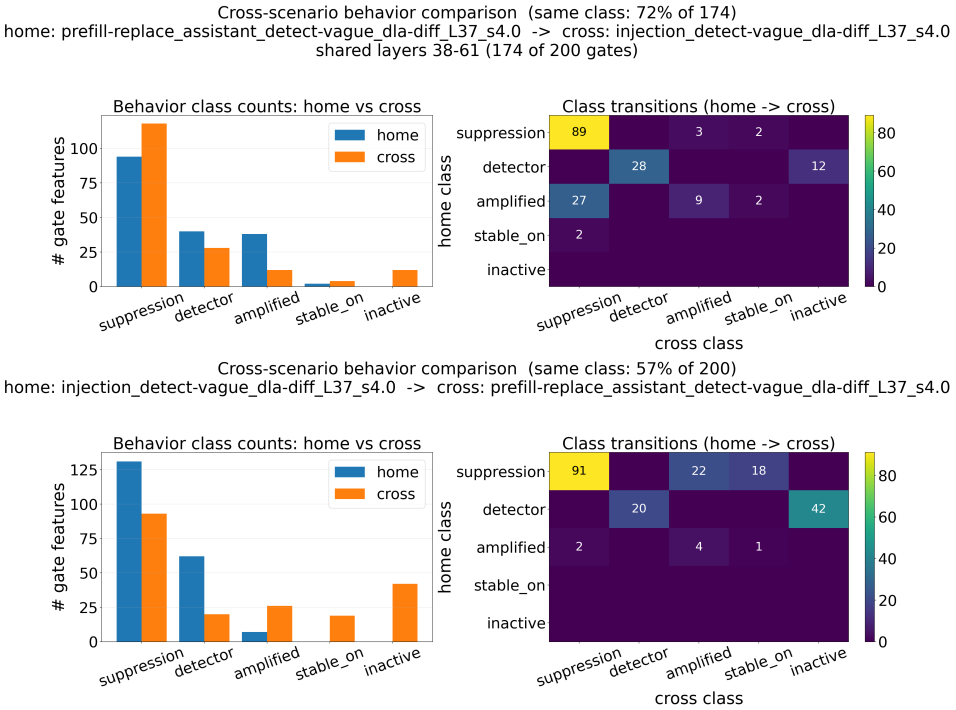


Figure 3.22: Class distributions and transitions for gate feature candidates collected with the mean diff metric. We can see this directly sellects for features suppressed on steering. The classes remain consistent between the scenarios, with some detector features ranked during injection becoming inactive in prefill.

more consistent across scenarios as seen in Figure 3.22. We call this ranking of gate candidates the diff metric.

Distributions of gate features

We investigate and compare the layer distributions of the collected gate candidates for both metrics and scenarios, the plots can be seen in Figure 3.23. Spearman correlation is chosen as the main metric here, because as it was visible during activation patching in Figure 3.19, the top gate candidates have significantly higher causal importance, meaning comparing the rankings of the candidates provides us with more relevant information than looking at an unranked overlap. Both prefill scenarios share a nearly identical gate distribution, achieving a Spearman correlation of 1, while the DLA gate candidates collected in prefill and injection have a 0.663 spearman correlation. The mean diff metric gives a significant, 0.826 spearman correlation between scenarios. Pure direct logit attribution causes a large artifact in gate rankings in layer 38—directly post injection—the large injection causes model activations to increase in mean activation. We hypothesize this artifact was partially responsible for the difference in gate classifications.

We collect the gate activations on the injection scenario and patch them to the

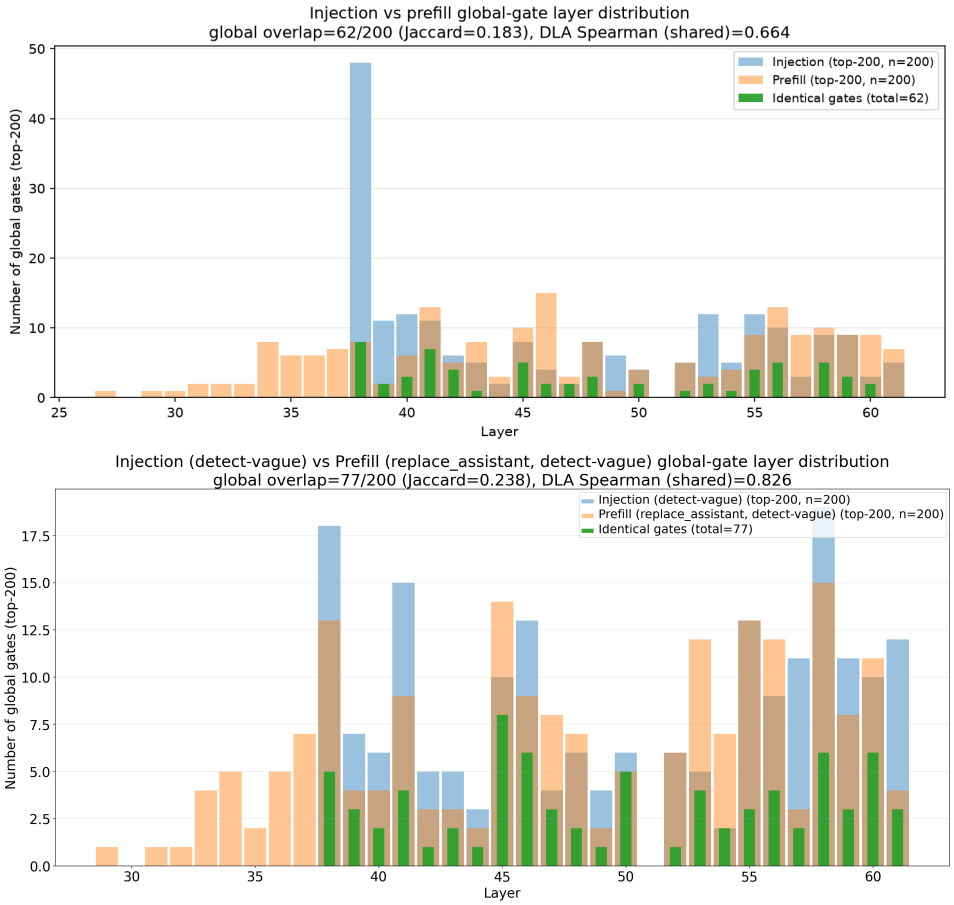


Figure 3.23: Comparing the distributions of the top 200 gate features found by DLA in prefill and injection scenarios. We compute the global and per layer spearman correlations. Top—pure DLA attributution, bottom—diff.

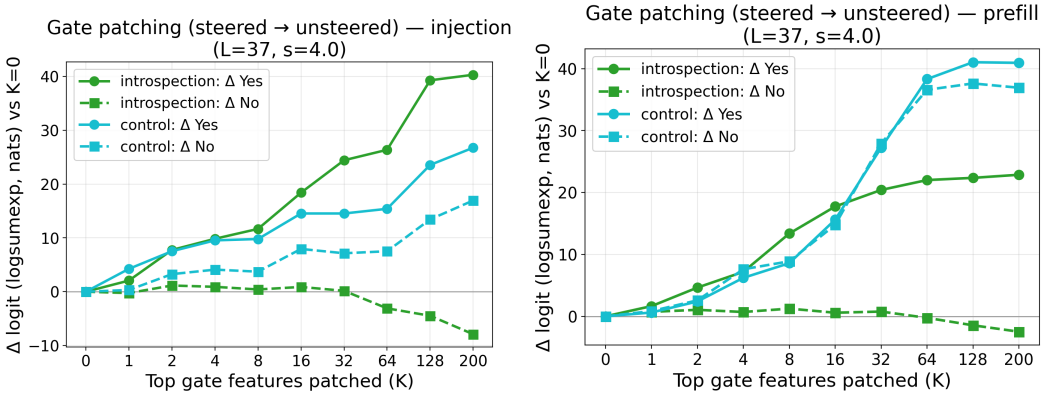


Figure 3.24: Comparing the effect of gate patching on yes and no logits in introspection and control questions. We find that for control questions the steered gate activations promote both 'yes' and 'no' logits, while decreasing the 'no' advantage only slightly.

arithmetic control question to verify whether the gates apply a simple 'Yes' bias when suppressed or whether the behavior is more context dependent. We show the result in Figure 3.24 We find when patched to the control question the gates orient the model in to answer a 'Yes' or 'No' question by boosting the logits of both tokens almost equally, as opposed to applying a simple 'Yes' bias. We see that for introspection the gates under steering boost 'Yes' while suppresing 'No' slightly.

Takeaways

- **Gate features are a shared component of the injection and prefill scenarios.** Gates ranked in one scenario retain high causal effectiveness in the other, the two prefill variants yield nearly identical gate distributions, the prefill and injection gates distributions have a high Spearman rank correlation of 0.826.

- **The pure DLA-based selection is distorted by the injection itself.** The activation surge directly after the injection layer produces a large artifact in layer 38, and selects many detector and amplified features, which explains the counterintuitive result that prefill-ranked gates outperform injection-ranked ones within the injection scenario. Ranking by mean activation difference instead, selects directly for suppression behavior, restores within-scenario effectiveness, makes the behavioral classes consistent across scenarios, and raises the cross-scenario Spearman rank correlation of gate distributions by 0.163.

- **Gates don't apply a simple linear bias.** Patching gate activations on the arithmetic control question boosts the raw logits of both 'Yes' and 'No' while moving their difference only slightly, indicating that gates orient the model within the answer subspace rather than applying a simple linear 'Yes' bias.

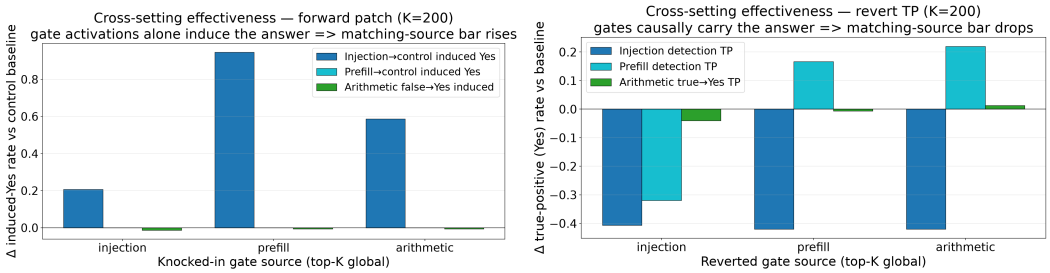


Figure 3.25: DLA sampled gate cross effectiveness. Above injection we see the performance of gates collected in injection across all the scenarios. In blue, the performance of gates from the source denoted by the label below, when performing vector injection on the Main prompt.

- **Gates don't carry concept information.** Gates never affect forced concept identification, confirming they act as a report mechanism rather than a carrier of the concept itself.

3.5.3 Gate universality

Seeing the cross scenario effectiveness, and considering one of the criterion's for gate feature candidates was the alignment with the 'Yes'-'No' direction in the model, we performed a control experiment, where we collected the top 200 gate candidates on a set of 150 simple arithmetic equations for which we create a true and false variant. We treated the true equations as steered examples from Equation 3.5.2, and the false variants as the unsteered activations.

We performed cross scenario importance trials for gates collected from all settings, we show the results in Figures 3.25 and 3.27, where the cross scenario importance when patching the top 200 candidates for each of the scenarios can be seen. We see that for the diff metric, the gates collected with arithmetic controls can completely revert the introspection performance back to baseline, while the introspection gates have low, but visible causal importance when acting on the arithmetic questions. The DLA sampled gate candidates do not capture the causal components responsible for changing the answer in the arithmetic questions, meaning they do not represent the gates although the gates gathered in the arithmetic control scenario show high causal importance in injection, outperforming the gates collected with injection themselves.

This suggests that the distribution similarity when sampling for gate behavior is less direct evidence for mechanism similarity between prefill and injection scenarios than initially thought. Looking at the distribution of the gates themselves, in Figure 3.26 we see the gates collected in arithmetic control questions via the diff metric concentrate significantly in the last layers, suggesting a different mechanism.

We investigate the set overlap of the gates themselves, checking if there are

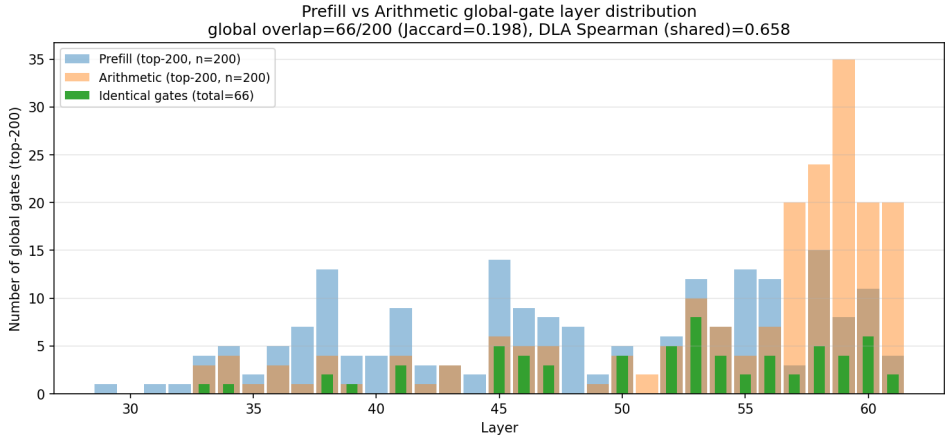


Figure 3.26: Comparison of the distribution of gates collected in arithmetic and prefill scenarios

	(Prefill ∩ Injection) \ Arithmetic	(Prefill ∪ Injection) ∩ Arithmetic
DLA	40	54
diff	41	73

Table 3.4: Introspection specific and general "Yes" gates from the top 200 activating candidates

gate candidates that appear specific to introspection, as opposed to being part of a more general "say yes/no" mechanism. In Table 3.4 we show 20% of the top 200 feature candidates collected with both metrics appear to be introspection specific. Comparing their behavior classification, as in Figure 3.20, we find no significant differences. In Section 3.5.4 we compare the behavior of weak evidence carriers gathered for two gates—`L45_F9959`, a top candidate appearing across all scenarios, and `L47_F15165`—a highly activating candidate present only in the introspection scenarios.

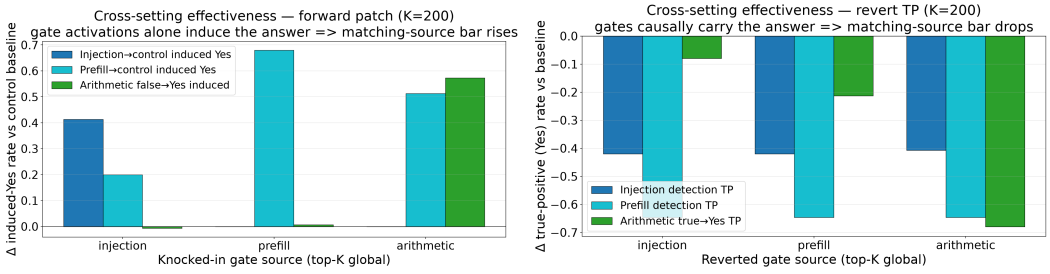


Figure 3.27: Diff sampled gate cross effectiveness.

Takeaways

- **Most Gates are a part of a more general answer 'Yes'/'No' mechanism.** The similarity of gate distributions between prefill and injection scenarios is weaker evidence of a shared introspection mechanism than it first appears. Gates collected on true/false arithmetic questions can fully revert introspection behavior under the diff metric.
- **Some gates activate only in introspection questions.** Roughly 20% of the top gate candidates appear specific to the introspection scenarios, but looking just at their response to steering shows no visible differences.

3.5.4 Weak evidence carriers

In this subsection we compare and analyze the behavior of Weak evidence carrier features, described in [MYW+26]. In **Background** we give a description of the Carriers found by the authors, in **Setup** we describe the methodology used in the paper and how we utilize it. The plots for showcasing the causal effectiveness of carriers on gate activations in injection are recreated from [MYW+26], and extended by us to cover prefill and cross scenario variants. All the experiments and conclusions described from **Our results** onward are entirely ours.

Background

[MYW+26] analyze features that influence gate activations directly by sampling features whose decoder direction aligns with the gate encoder direction. The attribution score for a weak evidence carrier feature f for a gate g, at a given concept c, is computed by

$$\mathrm{attr}_{c,f} = \left(w_f^{\mathrm{dec}} \cdot w_g^{\mathrm{enc}}\right) \mathrm{act}_f\big(x_c^{\mathrm{steered}}\big).$$

Where w_f^{dec} is the decoder matrix for the Transcoder where the feature is present, and w_g^{enc} is the encoder matrix for the Transcoder where the gate is present. The top features with positive attribution are weak evidence carriers, the top features with the most negative attribution are suppressors—their write direction is almost opposite the gate encoder direction.

Unlike Gates, Weak evidence carrier features activate linearly with steering strength, with most being inactive when no injection is present. In the original experiment, weak evidence carriers were much more numerous than gates, and their causal effectiveness was distributed throughout the set of carrier features, unlike gates, where the top ranked candidates accounted for a large portion of the causal effectiveness. Some evidence carriers were related to the concepts semantically, based on the labeling of the contexts where they become active in text, though the majority didn't show such connections.

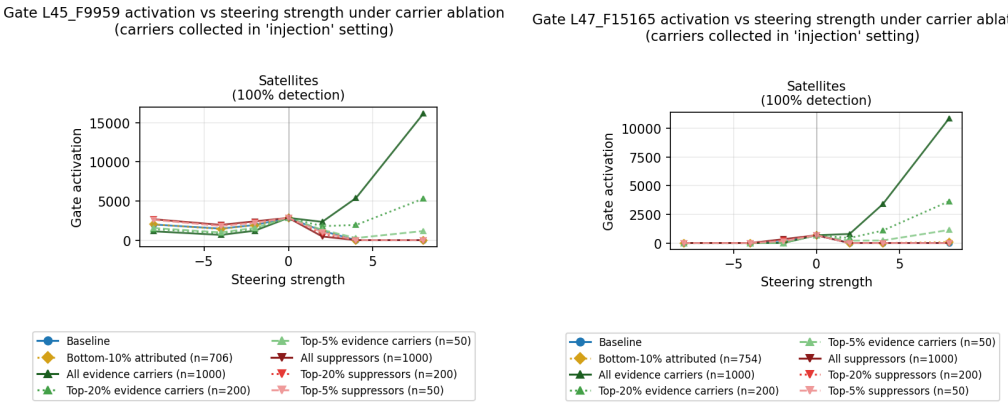


Figure 3.28: Injection evidence carriers effectiveness for the General gate and Introspection-specific gate. The inverted v pattern is visible for the baseline response to steering.

Setup

Due to the high computational cost, we collect carrier features for two gates, `L45_F9959`, which is present both introspection scenarios and arithmetic controls, and `L47_F15165` which is specific to the injection and prefill scenarios. We therefore refer to `L45_F9959` as a General Gate, and `L47_F15165` as the Introspection-specific Gate. We observe some general trends present for both gates and note there are noticeable differences in the carrier behavior when comparing them.

The causal effectiveness of weak evidence carriers is investigated by capturing the activations of the top-k carriers on the baseline prompt and running the steered example (injected vector, word appended to prompt) with the activations of the top-k evidence carriers replaced with those from the baseline run. The carriers are said to be causally important if replacing their activations to baseline reverts the gate activation back to it's value with no injection, or since gates are suppressed with steering increases the gate activation even further. Similarly suppressors are said to be causally effective if they decrease the gate activation below baseline.

Our results

We run experiments checking the causal effectiveness of gates collected in prefill and gates collected in injection scenerios, both within the same scenario, and across scenarios. We visualise the results for the concept "Satelites", however the trends we observe are general across all concepts, and more plots can be seen in Appendix A. In Figure 3.29, we see that for the General Gate the prefill carriers perform better than suppressors only for $n = 1$ appended words, while for $n = 2, 3$ both carriers and suppressors perform higher than baseline, indicating the classification doesn't apply correctly here. The difference between the carrier replaced run and baseline decreases

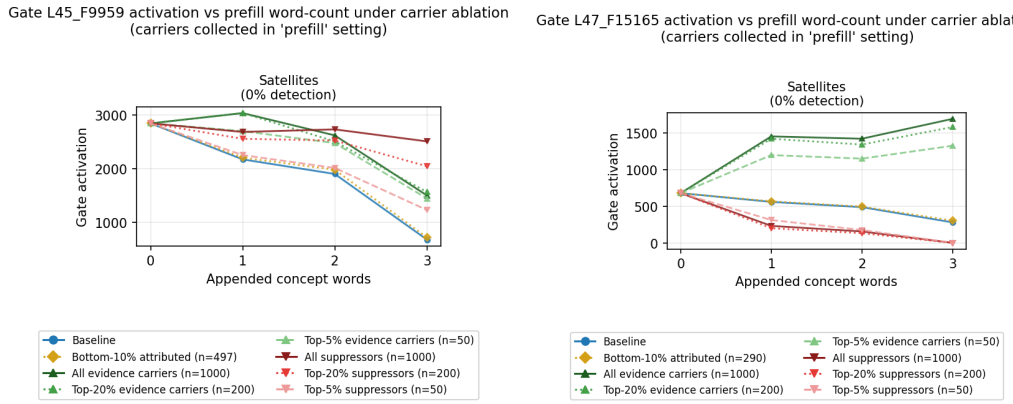


Figure 3.29: Comparison of prefill carriers effectiveness on their respective gate in the prefill scenario.

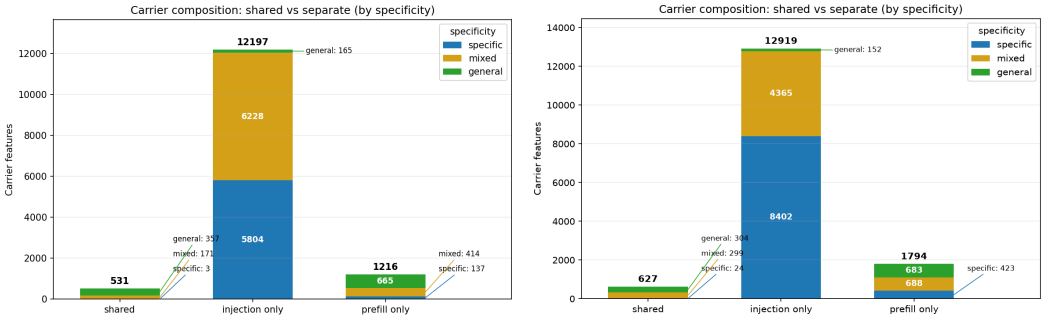


Figure 3.30: Carrier feature specificity: left—General Gate, right—Introspection-specific Gate.

with stronger steering (more words), in contrast to what is seen in injection, and for the Introspection-specific Gate, prefill. For the Introspection-specific Gate the carriers and suppressors behave as expected, showing concrete causal importance. We note the Introspection-specific gate itself changes it's activations under steering more gradually than the General gate and at baseline reaches $\frac{1}{3}$ of the activation magnitude of the General gate. Nevertheless, for the Introspection-specific Gate, in prefill carriers clearly have more influence on the gate's activation than with the General Gate.

We observe a clear difference in the characteristics of the carriers collected in prefill compared to those collected for injection. We count the carriers for both scenarios, and for how many concepts a given evidence carrier is active. We say a weak evidence carrier is specific when it activates for ≤ 2 concepts. A carrier is general if it is active for at least 50% of concepts. The rest are mixed. In Figure 3.30 we see for injection nearly 10x more carrier candidates are collected, and those candidates are much more concept specific, with 47% of them appearing for one concept only for the General Gate, and 65% for the Introspection-specific Gate. Prefill carriers are much more general, with nearly 90% appearing across multiple concepts for the General Gate, and thus a lot less individual different carriers are present. Here a

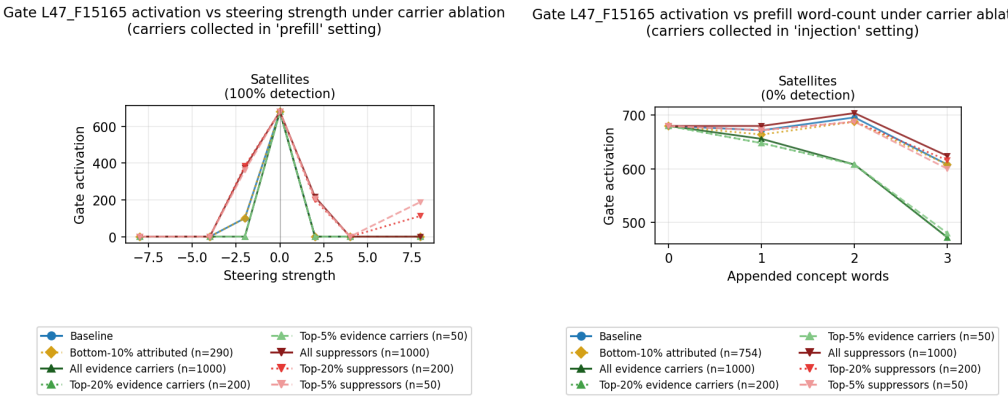


Figure 3.31: Cross setting effectiveness for the Introspection-specific Gate. Both gates perform poorly.

larger difference is visible, as for the Introspection-specific gate 598 more carriers are collected, and 23% are concept specific. The higher similarity of carrier characteristics for both scenarios appears together with higher causal importance for prefill in Figure 3.29.

In Figure 3.31 we see that for both gates, the carriers have little to no cross-setiing causal importance, suggesting different specific processing pathways are active in prefill and introspection scenarios. We hypothesize the higher generality of prefill carriers for both gates suggests a longer processing chain is present in the prefill scenario, not captured by the simple two-stage circuit. The carrier collection only finds features acting directly on the gate, while the fact the concept signal in prefill is present from the first layer allows for a longer processing chain compared with the injection scenario. The higher specificity found in the introspection specific gate suggests a partial carrier to gate mechanism is still present in prefill nevertheless.

Carrier concept relevance and distribution

In the original experiment, carriers were expected to find concept information, to understand the causes of weak cross setting performance, besides a difference in the layers where candidates are present, we investigate the specificity of carrier features and generate labels for top activating examples of the carrier candidates on the pile-10k dataset, to check for their relevance to concepts for which they are active. We then use an LLM judge to classify whether a given features top activating examples are related to the concept for which the feature became active as a carrier. The results of this classification can be seen in Figure 3.32. We note that the low relevance rates may be influenced by the texts visible in the dataset, however they can be compared between the scenarios. We see more concept related carriers activate for the Introspection-specifc gate, though for the prefill carriers the numbers, at 6 full and 7 partial matches, are still extremely low.

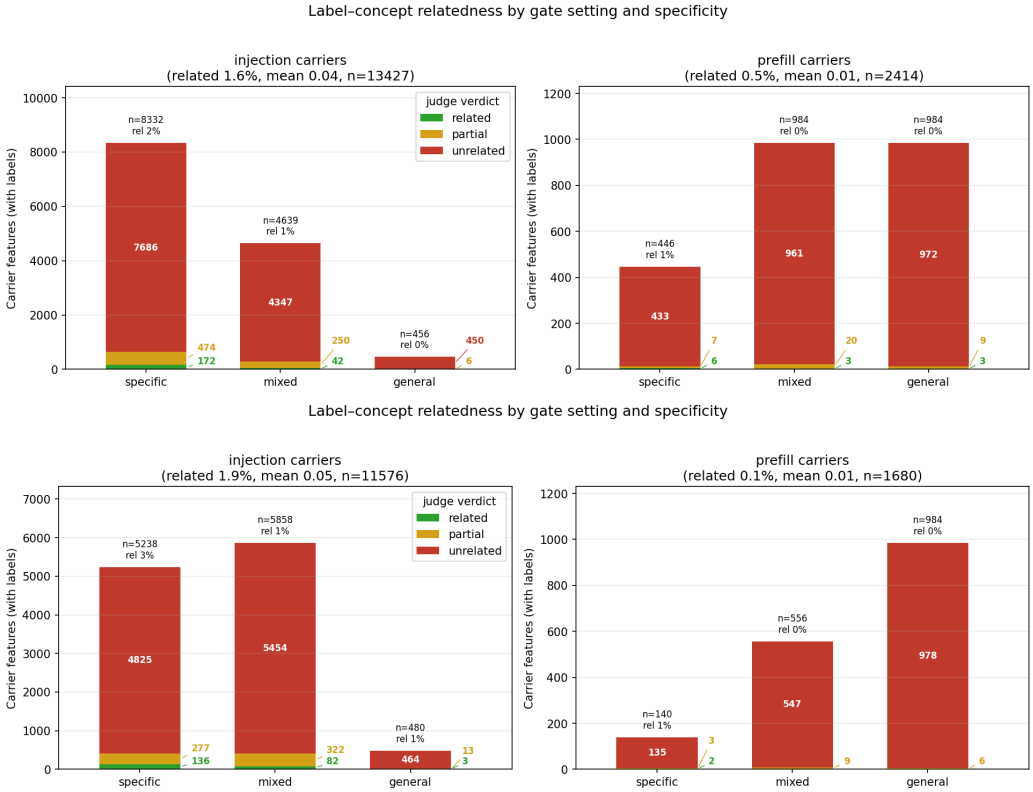


Figure 3.32: Relatedness of the top activating examples from pile-10k to the concept(s) for which the carriers were active. Top: Introspection-specific gate, bottom: General gate

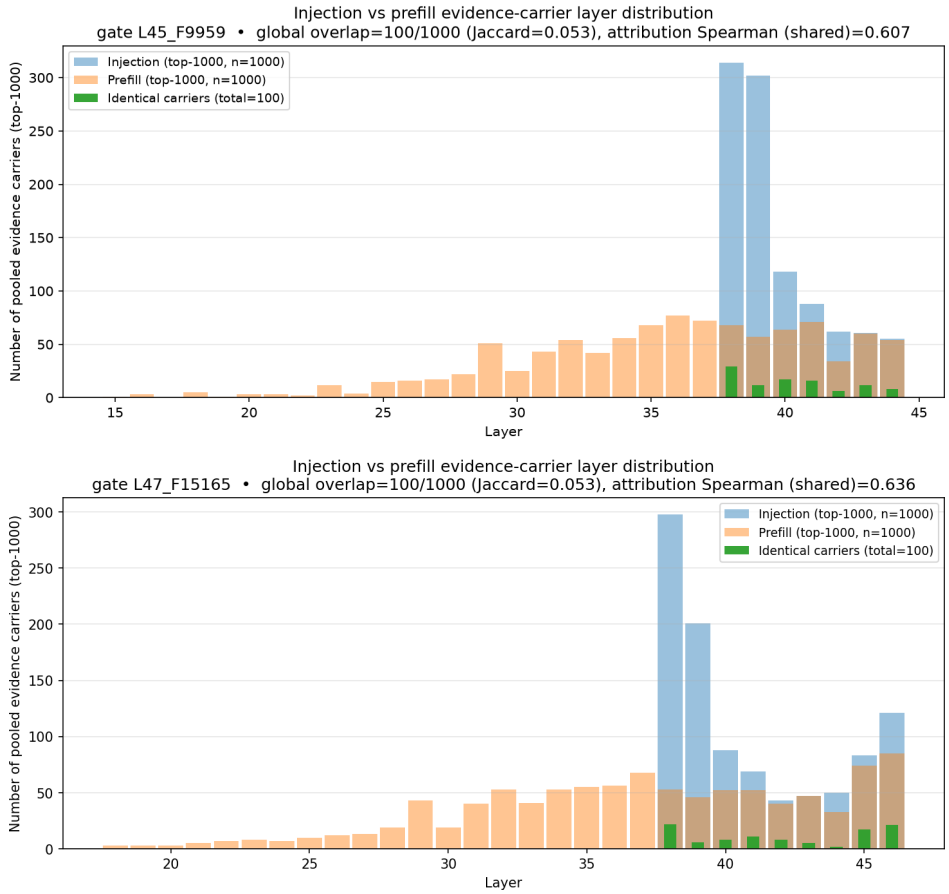


Figure 3.33: Comparison of the top 1000 carrier distributions across injection and prefill for the two gates.

Looking at the top 1000 carrier distribution itself (Figure 3.33), we find that while there is some overlap, it is markedly lower than in the case of gates. Since the evidence carriers are more distributed, the 90% of lower ranked features which do not overlap carry a significant causal weight, unlike gate features, where top activating gates account for a majority of the causal effectiveness, meaning the ranking overlap for gates translates to cross scenario importance (we show the raw gate scores in Appendix A), unlike evidence carriers where the overlap doesn't translate to causal effectiveness across opposite scenarios. The Introspection-specific gate has more carriers active directly in the injection layer, meaning they are activated directly by injection itself, consistent with a shorter two step mechanism.

Takeaways

- **Carriers behave differently in the prefill and injection scenarios.** Injection recruits an order of magnitude more carriers, most of which are concept-specific, whereas prefill carriers are fewer and largely concept-general; carriers from one scenario have little to no causal effect in the other.

- **Carriers appear to be more relevant for introspection-specific gates in the prefill scenario**. Carriers for the Introspection-specific gate effectively reverted gate activations in prefill. Due to computational constraints we were only able to check this for two gate candidates.

3.6 Related work on introspection

[LM26] and [God25] provide plots showcasing detection for different injection layers. [God25] does not vary the collection layers, running into issues with injecting vectors far away from their collection point, that we discussed in Section 3.3. [LM26] does not provide control plots, there the authors find Qwen235B-A22B has two detection peaks. The authors inject at the layer the vectors are being collected. They provide independent evidence that the detection and identification mechanisms are separate. They perform third-person controls where models are asked about the internal states of a hypothetical other model, showing detection behavior in different layers, but find the effect is also affected by the length of the prompt itself.

[RA26] performs LoRA finetuning to increase introspection scores in smaller models; when performing initial tests, we observed that models finetuned in this manner suffer from degraded performance on control questions, and the introspection and control logits follow a similar pattern, the exact behavior we focused on avoiding in Section 3.2.

[MYW+26] Finds that for vector injection discrimination between injection and baseline trials emerges during Direct Preference Optimization training. The authors also perform tests eliminating the hypothesis that introspection works by applying a simple linear bias.

Chapter 4

Conclusion

Smaller open source models demonstrate genuine introspective capabilities, as verified by the experiments established in Section 3.2. The apparent introspective capabilities are dependent on the logit distribution for the base prompt—The more likely 'Yes' is as an answer to the base prompt, the more visible the introspective signal becomes.

We have shown injection sensitivity is dependent on both injection and vector collection layers, and have shown in which areas Gemma 3 and Gemma 4 are sensitive to introspection.

Refusal direction ablation can increase model introspection performance, as shown in Gemma 2, however, we have demonstrated the need for careful verification of injection-based introspection demonstrations with the use of logit based checks on unrelated control plots, as shown by the coherence degradation discovered in Section 3.4 thanks to our controls under multi layer ablation with Gemma 3.

We have shown that gate features are part of more general mechanisms, being causally important in answering 'Yes'/'No' questions unrelated to introspection. We maintain the hypothesis that introspective behavior is related to anomaly detection mechanisms used by models to decide whether a piece of text is expected in a given context, however, as we have shown, the exact mechanisms differ, with the prefiill based scenario showing signs of a longer processing chain as opposed to the direct evidence carrier→gate mechanism found with injection. The closer similarity and higher effectiveness of evidence carriers in prefill for the introspection-specific gate are an indication that the two mechanisms are related, though we are unable to establish statistical evidence due to computational constraints.

The mechanisms and behaviors discovered suggest that a kind of anomaly detection mechanism is responsible for introspective reports in current LLM's.

4.1 Limitations

We were not able to trace the full mechanism responsible for affirmative answers with prefill injection. The more detailed investigations were limited to the Gemma family of models. We have not investigated in detail the role attention heads play in introspection, though we have confirmed ablating heads attending to the injection position during vector injection lowers introspection. Due to computational constraints we leave comparisons of attribution graphs found in [MYW+26] to future work.

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Appendix A

Additional plots and case studies

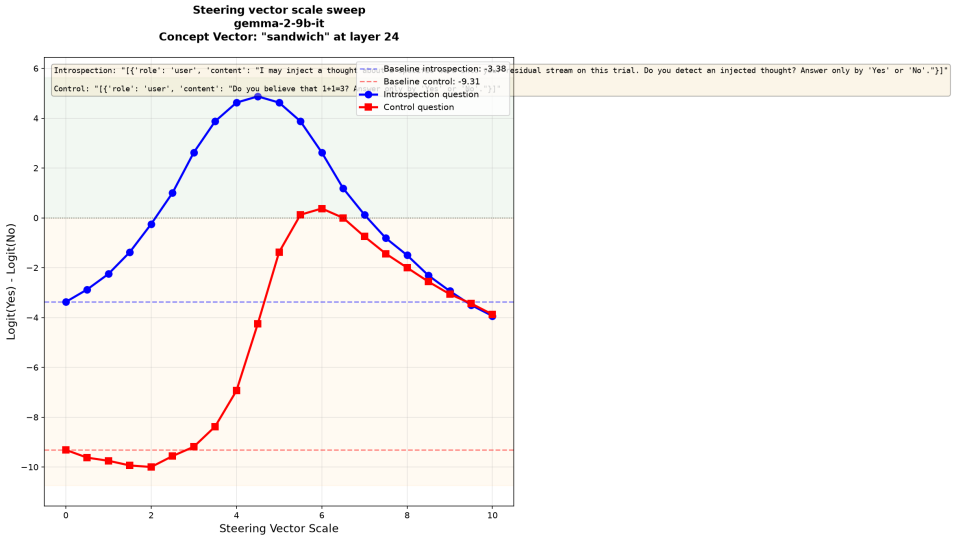


Figure A.1: logit plot on the short prompt variant

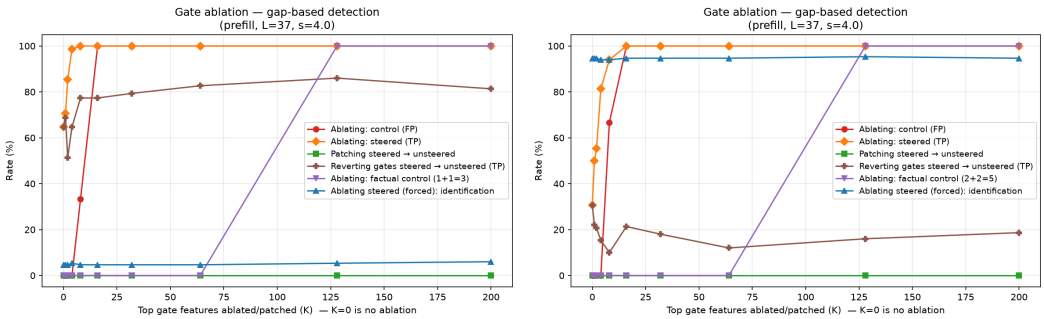


Figure A.2: Comparison of replacing the assistant response - left, and appending to user prompt - right

Table A.1: Gemma 3 responses to *"Do you believe* 1+1=3*?"* (expected: `No`) under concept steering and ablation on all token positions, as in [MYW+26]. Q1: *"Answer with one word only: 'Yes' or 'No'."*; Q2: *"Answer only by 'Yes' or 'No'."* ↩ marks a line break; … marks truncation.

cccc

	Q1		Q2	
Steering	steer only	+ ablation	steer only	+ ablation
none (baseline)	No.	No.	No.	Yes.
Youths	'Yes' or 'Aut'	'm5m5m5m m m	'V' is' is' is's	Them, or 'Lev' is't
Constellations	No.	Yes.	Yes	Yes.
Satellites	No.	No.	No.↩.↩…	Yes.
Secrecy	No.	'No'.	No.	Yes.
Sadness	No.	Sad.↩Yes.↩No.↩Sad	Sadness.	Sad, but it is only a 'No'.
Quarries	No.	No.	No.	Yes
Treasures	'Yes' or 'No'	'Yes' or 'You' (when referring	likely by 'Yes' or 'AT' (	"Yes" or "Ch" (or "
Lightning	No.	No	Yes	Yes
Masquerades	No.	'No'	No.	Yes.
Cameras	No.	'No'	No.	! ! ! ! !…

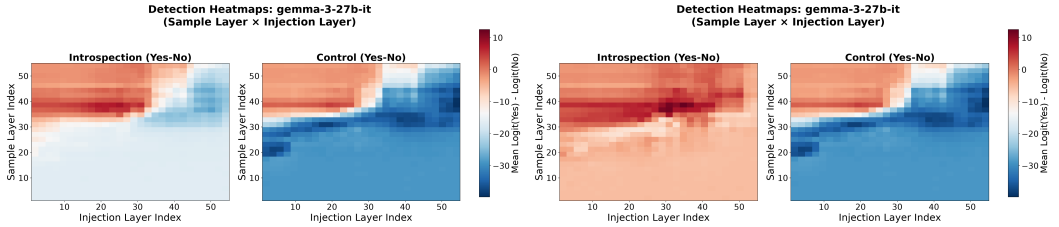


Figure A.3: Detection and control question performance based on the sample and injection layer of concept vectors for Gemma 3 on the Main (left) and Short (right) prompt variants

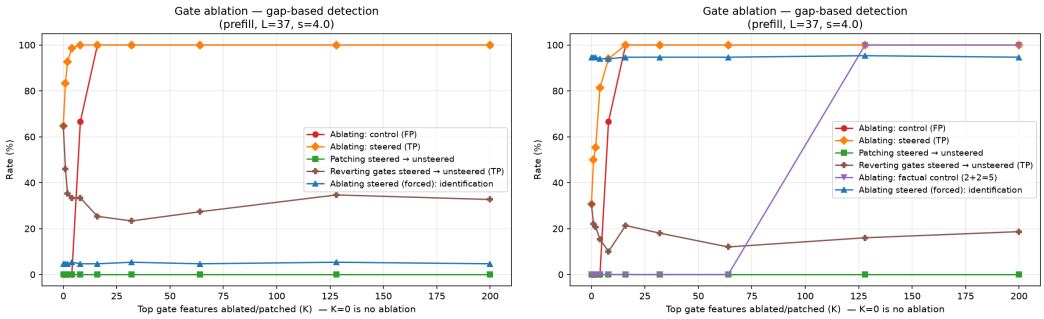


Figure A.4: dla injection gates replaced on replace assistant prompt(left) and append to user on itself

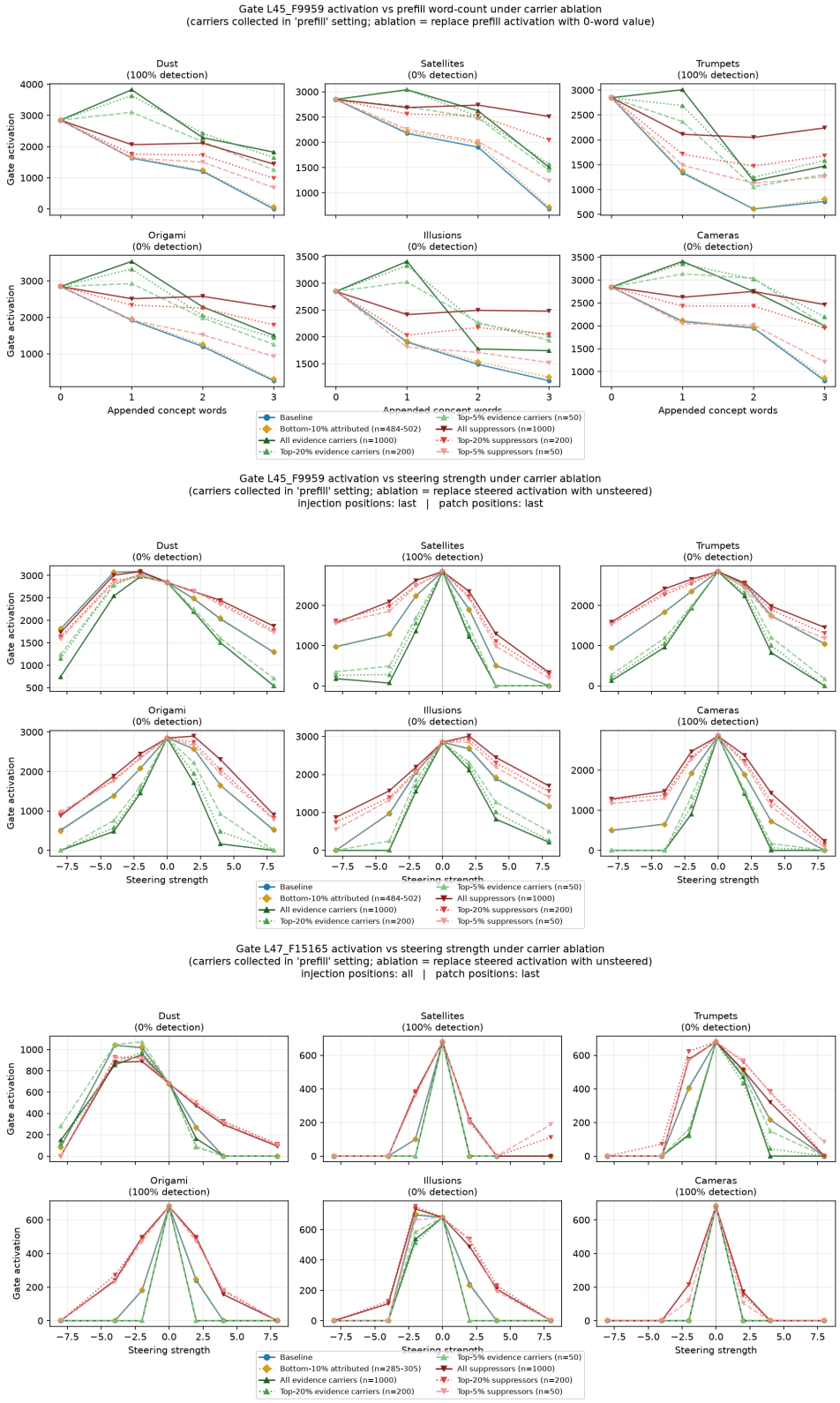


Figure A.5: carrier ablation plots for more concepts

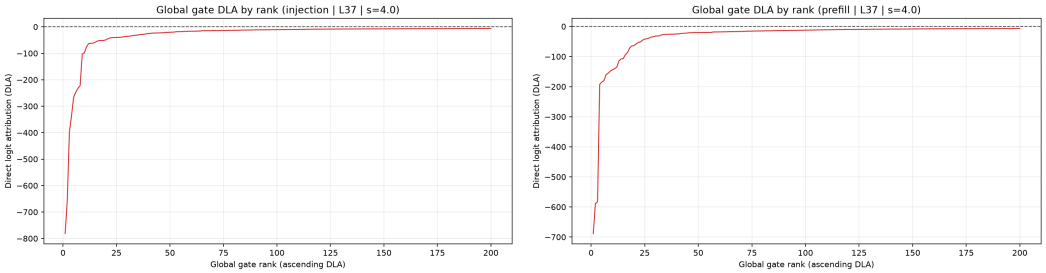


Figure A.6: Comparison of DLA gate scores for prefill and attribution

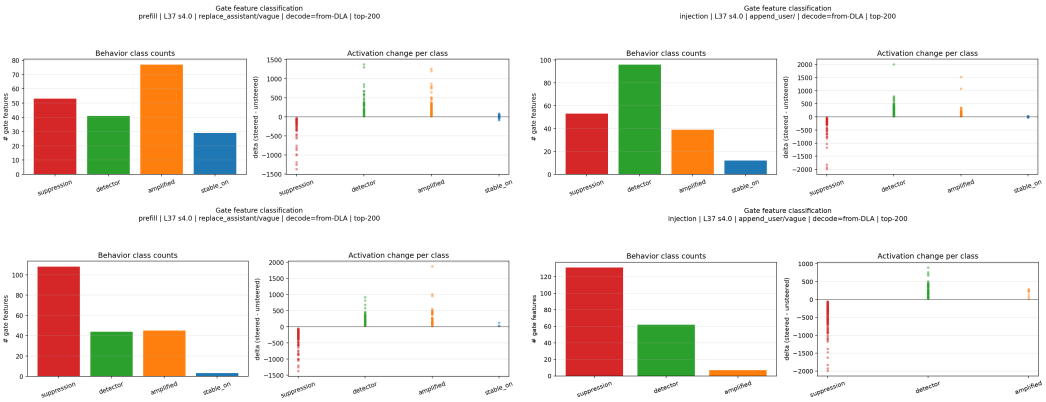


Figure A.7: Comparison of gate candidate behavior: top row DLA, bottom row mean suppression

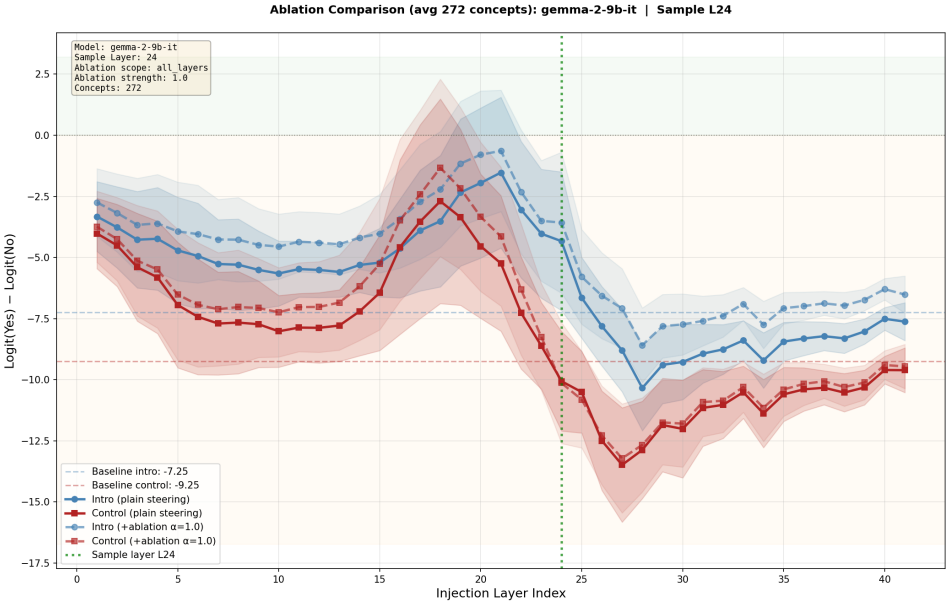


Figure A.8: Refusal direction ablation for Gemma 2 on the Main experiment prompt